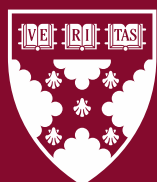


Working Paper 26-090

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# AI-Native Firms\*

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We study how firms built around AI capabilities—“AI-native” firms—are organized. Drawing on Y Combinator batches W20–F24 and U.S. venture-backed startups whose first financing closed between 2020 and 2024, we classify each firm’s AI-native status and link it to workforce microdata on team size, function, seniority, and hierarchy. Relative to non-AI startups in the same industry-cohort, AI-native firms are 25% smaller. Their share of engineers is 13% greater, and the shares of entry-level workers and managers are each roughly 15% lower. Their hierarchies are half a seniority level flatter—yet valuations are comparable, suggesting higher value created per employee. We argue these patterns reflect two channels: a *process* channel, in which AI changes how people work inside the firm, and a *product* channel, in which AI capabilities are built into what the firm sells. Using text from product descriptions and job postings, we find that embedding AI into the product, beyond layering on AI tools into existing workflows, is a primary way startups are scaling “knowledge work” without large teams of knowledge workers.

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# 1 Introduction

Technology reshapes organizational form. Railroads and the telegraph allowed firms to serve national markets, increasing the returns to coordinating across distance and creating the conditions for large managerial hierarchies (Chandler, 1993). Electricity altered how work could be organized on the factory floor and spurred the rise of scientific management (David, 1990). The internet lowered communication and coordination costs, supporting more platform-based, networked, and distributed forms of organization (Shapiro and Varian, 1999). Across technologies—from lithography to Linux—product and organizational architectures often co-evolve (Henderson and Clark, 1990; Colfer and Baldwin, 2016).

How will AI reshape firms? Recent research has largely focused on a *process* channel, emphasizing how AI tools augment or automate work inside the firm through agentic coding, AI-assisted customer service, automated sales operations, and many other internal tasks (Peng et al., 2023; Brynjolfsson, Li, and Raymond, 2023; Dell’Acqua et al., 2023; Csaszar, Ketkar, and Kim, 2024; Otis et al., 2025; Sarkar, 2026; Kim, Kim, and Koning, 2026). Yet AI can be used for more than internal work. AI-native firms embed AI into their products, allowing customers to generate slide decks with AI, build software, and even run their accounts receivable. This *product* channel shifts productive capability from the internal organization into the product, moving knowledge work that would have required large teams to be performed directly in the product itself through imported model capabilities. Both channels suggest that “AI-native” firms—those that are built around AI-enabled products and processes—may be organized differently than comparable non-AI firms.

In this paper, we highlight this product channel and explore whether AI-native firms are, in fact, organized differently. We focus on startups because new firms are less constrained by legacy structures, making them a likely setting in which new organizational forms first become visible. Measuring whether a firm is AI-native presents a challenge because, as our two channels illustrate, there are many ways companies can be built around AI. Here, we leverage the fact that Y Combinator (YC), a leading accelerator for early-stage companies, curates and encourages startups to tag themselves with public labels to help investors, journalists, and customers discover new ventures they might be interested in. We leverage the fact that startups can add an “AI” tag to their public profile, enabling us to identify startups that claim that AI is central to their firm, whether that is embedding it into their products, using it as a core part of their production process, or something else entirely. To test if our YC results generalize beyond this extremely selected set of firms, we also

use text describing each YC startup to train a model that predicts any startup’s AI status, allowing us to extend our analysis to the broader universe of venture-backed U.S. startups in PitchBook. We then link both samples to Revelio Labs for detailed workforce data on team size, hierarchy, seniority, and functional composition.

We structure our analysis into two parts. First, we document a robust set of organizational differences between AI-tagged and non-AI-tagged startups. We find that AI-tagged startups have approximately 25% fewer employees than non-AI firms in the same industry-cohort cell. Their workforces are also more technical: the share of engineers is 5 percentage points higher, offset by lower shares in sales, finance, operations, and administration. They skew senior, with the share of entry-level workers just under 15% lower and the share of senior workers about 20% higher. Their hierarchies are flatter: half a level less depth and 15% fewer managers, even controlling for firm size. These patterns are consistent across our YC and PitchBook samples.

These smaller teams raise similar amounts of funding and hit valuations on par with non-AI-native firms, suggesting that they are not lower quality entrants. As a consequence of being smaller, AI-tagged startups raise roughly 20% more capital per employee and carry higher valuations per employee. These smaller teams also draw from a narrower labor pool: they are geographically concentrated in Silicon Valley, and employ workforces that are more male, more likely to hold advanced degrees, and drawn from more prestigious employers and institutions. Our findings are consistent with AI complementing technical expertise at the firm level, with small teams of skilled founders building ventures that historically might have required much larger hierarchical organizations.

The second part of our analysis explores the role of the *product* and *process* channels in explaining these patterns. Focusing on our YC sample, for which we have richer text data, we use an LLM to inductively classify founders’ own descriptions of their businesses and ask how AI is embedded into the product. 43% of AI-tagged YC startups use AI to fully automate tasks workers used to do, 24% build AI tools designed to augment existing workers at the firms they sell to, and 15% build AI infrastructure, providing a capability that another developer integrates into their own AI product. To measure the process channel, we code whether each firm’s job postings name specific worker-facing AI tools like ChatGPT, GitHub Copilot, or Cursor. AI-native startups are 2.6 times as likely as non-AI startups to name such tools in their postings, but this measure does not predict smaller firm sizes in the log, Poisson, or hierarchy specifications. Even when controlling for this process channel, we find that embedding AI into a startup’s products is associated with less headcount and flatter structures.

If embedding AI into the product moves productive capability from the firm’s workers into the product itself, then the firms that should shrink most are those that, without AI, would have scaled by hiring people to deliver the underlying work—therapy, tutoring, telemedicine, and similar service businesses where small teams can now build AI capable of doing the work that had required building larger workforces. To test this idea, we test for heterogeneity in our estimates by whether the startup is classified as a “services” business. The difference between AI and non-AI firms is significantly larger for these services firms: AI startups are roughly 70% smaller than non-AI peers and have hierarchies nearly a full level flatter.

Our paper makes three contributions. First, we identify a product channel largely overlooked in recent work on AI’s productivity impacts. The dominant research paradigm today focuses on the process channel: how AI tools change the firm’s internal workflows (Brynjolfsson, Li, and Raymond, 2023; Dell’Acqua et al., 2023; Kim et al., 2024). But a key way in which firms leverage AI is to embed it directly into what they sell, so that model capabilities perform, augment, or coordinate knowledge work through the product itself. This matters because it changes where the firm’s productive capability resides. A long line of work models the firm as a system for processing information and knowledge (Galbraith, 1973; March and Simon, 1958; Simon, 1947; Garicano, 2000), and firm performance has been shown to depend on how organizations access and act on decision-relevant information (Kim, 2025). Our findings suggest that AI enables firms to shift some of this processing from the organization into the product itself, allowing the firm to deliver knowledge-work capabilities without building them internally through people and hierarchy (Ide and Talamás, 2025). If this pattern holds, then building, importing, and orchestrating model capabilities may matter as much for competitive advantage as building human ones (Nelson, 1985; Teece, Pisano, and Shuen, 1997; Helfat and Peteraf, 2003).

Second, we show that these AI-tagged firms employ smaller teams of more talented and technical workers. These workers are especially likely to be graduates from elite institutions, concentrated in Silicon Valley, and male. While at the task-level AI often helps less experienced workers perform better (see Brynjolfsson, Li, and Raymond, 2023; Noy and Zhang, 2023; Csaszar, Ketkar, and Kim, 2024, ; cf. Otis et al. 2025), at the firm level we see a shift broadly in the opposite direction. Existing work shows generative AI changes labor demand when adopted by existing firms (e.g., Gupta et al., 2025; Brynjolfsson, Chandar, and Chen, 2025); we show that AI also changes the kind of firms that are created. While this may have implications for career opportunities and labor market outcomes (Beane, 2024; Kang and Kim, 2025), we urge caution in moving from firm-level to market-level

estimates. While AI-tagged startups employ fewer workers, they appear much less costly to launch and scale. If AI leads to meaningfully more and faster growing startups (Gupta et al., 2025), then while each firm may be smaller the overall demand for labor could well rise.

Third, our finding on the product and process channels is consistent with emerging evidence on the “mapping problem” in AI adoption. Kim, Kim, and Koning (2026)’s field experiment shows that the binding constraint to AI adoption is not access to the technology but the ability to search across a firm’s own processes and discover where AI can create value. Firms that solve this problem grow faster while requiring less external capital and no additional labor. Our null process-channel result, though a coarse and noisy proxy, is consistent with this view: equipping workers with ChatGPT, Copilot, or Cursor does not, on its own, predict smaller firms. Capturing real value from AI often requires re-engineering processes around it—and, as our results suggest, re-engineering the product itself so that AI does the work directly. The AI-native ventures in our data appear to have solved this problem by building the firm around AI from the start. More broadly, our findings suggest that when firms are fully built around AI, the technology does more than lower the cost of entry (Ewens, Nanda, and Rhodes-Kropf, 2018; Reimers and Waldfogel, 2026), it changes how ventures are organized and scale.

The remainder of the paper proceeds as follows. Section 2 elaborates on what makes a firm AI-native, distinguishing the process and product channels. Section 3 describes our two complementary datasets—Y Combinator startups and the broader universe of U.S. venture-backed firms from PitchBook—along with workforce data from Revelio Labs. Section 4 presents results on firm size, functional composition, seniority, hierarchy, and capital efficiency. Section 5 then explores the role of the product and process channels in explaining these results. Finally, section 6 discusses implications and limitations.

## 2 What Makes a Firm AI-Native?

The term “AI-native” firms has emerged in industry discussions to refer broadly to firms built around AI capabilities. Industry reports suggest that AI startups may be growing with smaller teams, but provide widely varying estimates and limited insight into how organizational form is changing (Tables A1 and A2). One reason may be that the term “AI-native” is used to describe several distinct ways in which firms rely on AI. In some accounts, AI-native firms are those that use AI as an internal production technology or core operating model, changing how employees code, sell,

design, or coordinate work. We refer to this as the *process* channel. In other accounts, AI-native firms are those that produce products and services that would not have been possible without AI, ranging from foundation model providers to companies that embed AI directly into what they sell. We refer to this as the *product* channel.

The process channel has been the focus of most existing research, which has largely focused on how AI can either augment or automate tasks within firms. The majority of recent research on the impact of generative AI on workers and firms has focused on this channel, exploring how AI changes the productivity of workers on individual tasks like coding, customer service, and strategy development (Peng et al., 2023; Brynjolfsson, Li, and Raymond, 2023; Csaszar, Ketkar, and Kim, 2024). This work has sparked speculation about a process channel by which organizations can succeed with smaller teams equipped with AI tools that enable workers to match the productivity of much larger teams. Building on these studies exploring the impact of AI on individual tasks, recent experimental evidence suggests that AI can have firm-level impacts on performance, growth, and productivity, but only when firms know how to map it into production (Kim, Kim, and Koning, 2026).

Alongside such internal use, firms are also increasingly embedding AI into their products. The organizational implications of this *product* channel are not immediate: adding AI features to a product may leave the firm’s structure largely unchanged, much as shifting from one web development language to another is unlikely to alter the firm’s underlying structure or production function. The implications are different, however, if AI allows the product itself to perform knowledge work that would otherwise require human teams. In this case, capabilities that once had to be built inside the organization can be embedded directly into what the firm sells.

A useful way to understand this is to start from the view of the firm as an information-processing system. Firms develop routines and hierarchy to help workers process information and coordinate effectively (Simon, 1947; March and Simon, 1958; Galbraith, 1973; Garicano, 2000). These structures make firm growth possible, but they also tie growth to a human constraint: knowledge must be allocated and processed through people and hierarchy (Garicano and Rossi-Hansberg, 2015), decisions require judgment under uncertainty (Agrawal, Gans, and Goldfarb, 2022), and organizational performance depends on the scarce attention and coordination needed to translate decisions into action (Dessein, Lo, and Minami, 2022; Kim, 2025). Growth therefore typically requires expanding the human organization through which knowledge work is performed.

When AI is embedded in the product, some of this knowledge work can be performed directly



through what the firm sells. Producing personalized recommendations for different types of customers no longer requires a team to segment users, tailor messages, and coordinate around the task, since AI can perform this work inside the product. Providing credit score estimates from within the product does not require a division of workers producing such predictions. AI thus enables capabilities that once required repeated human workflows inside the organization to move into the product, potentially changing how firms scale and compete (Nelson, 1985; Teece, Pisano, and Shuen, 1997; Helfat and Peteraf, 2003).

This argument builds on Ide and Talamás (2025)’s formal model of AI’s impact on knowledge hierarchies within firms. Building on hierarchical models of knowledge work (Garicano, 2000), they model AI as making noncodifiable expertise scale through compute rather than relying only on scarce human time and knowledge. They distinguish between nonautonomous AI, which advises human workers, and autonomous AI, which can pursue production opportunities independently. In their model, autonomous AI changes production because routine work can be performed by AI agents rather than human workers, reallocating humans across production and problem-solving roles and giving high-knowledge workers greater leverage.

Building on this insight, we focus on what happens when autonomous AI is embedded in the product rather than deployed inside the firm’s production process. Scaling outputs through knowledge work traditionally requires organizing more production internally: employees scope requests, coordinate on work, and deliver results. Autonomous AI can make each workflow less labor-intensive and shift employees toward higher-level tasks, but producing more outputs still requires the firm to manage more work. In contrast, when AI is embedded in a product, customers can generate outputs directly through the product. Users specify the task, interact with the AI, evaluate and revise results, and iterate directly in the interface. While the firm still invests in building the product, output growth can be scaled through compute rather than by expanding internal teams.

This shift may have implications for how firms are organized. First, it may lower the need for human labor per unit of output, because serving additional customers does not require a proportional increase in internal production work. Second, it may contribute to flatter structures, because customer requests can be handled through the product rather than routed through a knowledge hierarchy. Third, it may increase technical talent density, as larger shares of employees focus on building and improving the product rather than carrying out routine production tasks. More broadly, productizing capabilities that would have otherwise been delivered through internal workflows can change how much labor firms need, as well as what kind of people and organizational design they

need to scale.

Recent cases illustrate how building around AI may shift how firms are organized. Gamma, an AI presentation startup, builds on image and text foundation models to generate complete slide decks for its users (Koning, Keller, and Kim, 2025). While Gamma’s engineers use AI tools extensively, including tools such as Cursor for software development, the case illustrates how the largest organizational implication comes from embedding AI into the product itself. One way to view Gamma’s product is as the deployment of millions of autonomous or semi-autonomous AI agents, each pursuing a production opportunity on behalf of a user: generating layouts, writing copy, selecting images, and localizing content across dozens of languages. In contrast, a traditional presentation service company, even one using AI internally, would still scale by turning each deck request into an internal workflow: scoping the request, coordinating design and copy work, and reviewing revisions before delivering the final deck. Gamma instead turns the request into a product interaction. The firm still needs employees to build and improve the product, but it does not need to organize a human workflow around each deck. As a result, a team of roughly 30 employees could serve millions of users and reach \$50 million in annual revenue in just two years.

FazeShift shows how the same product channel logic operates in the enterprise sector (Koning, Kim, and Keller, 2025). The company builds AI agents that automate the full accounts receivable lifecycle, including invoicing, collections, and cash application—work that previously required dedicated AR teams at each customer. With a team of just ten, FazeShift landed dozens of enterprise customers by building agents that match payments to invoices, send context-aware collection messages, and resolve discrepancies that human analysts used to handle manually. A traditional AR SaaS vendor would need a much larger team both to build out the firm-specific exception handling for the AR product and to staff the operations to resolve the exceptions that FazeShift’s AI agents now handle automatically.

Taken together, these arguments suggest AI-native firms should look different. On the process side, we should expect such firms to be more productive, with smaller teams able to do the same amount of work—although if AI dramatically lowers the cost of what the firm sells, a Jevons-style expansion in demand could lead firms to hire more rather than fewer workers. On the product side, as firms move AI capabilities into what they sell, they may require different sorts of talent: workers with the technical expertise to embed AI into a product and to complement, rather than be replaced by, those capabilities. Both channels suggest that AI-native firms should be organized differently from their non-AI peers, as the technology changes both how the firm works internally and what

the firm needs to build. In the next section, we turn to the data to ask whether AI-native firms do in fact look different, and how each channel—process and product—account for the differences we find.

### 3 Data and Research Design

We construct two complementary datasets to examine this question. Our primary dataset focuses on startups that attend YC, which provides us with clean measurement of AI status and startup age. Our secondary dataset is PitchBook, which allows us to extend our analysis to the broader universe of venture-backed startups in the U.S. to test the external validity and generalizability of our core YC estimates.

#### 3.1 Y Combinator Sample

We construct a dataset comprising all YC startups that participated in batches from Winter 2020 to Fall 2024, collected through an open-source public API.<sup>1</sup> Since YC publishes complete rosters for each cohort, it offers a well-defined sample through which hundreds of young but promising companies pass each year. The 2020–2024 window spans GPT-3’s public release in 2020 and ChatGPT’s breakout in 2022—the period when AI funding and startup formation surged. Our initial sample contains 2,891 firms across 11 YC cohorts (W20, S20, W21, S21, W22, S22, W23, S23, W24, S24, F24).

##### 3.1.1 AI Classification

YC encourages startups to tag themselves with descriptive labels to facilitate discovery by journalists, potential users, and investors. A startup like FazeShift, an “AI agent for Accounts Receivable,” is marked with tags including “artificial-intelligence,” “fintech,” and “b2b.” We classify a startup as an AI startup if its tags include “artificial-intelligence” or “ai.” Reflecting the general-purpose nature of the technology, AI startups build around AI in varied ways: from autonomous agents to AI driven drug discovery to AI-guided physical work. Self-identification lets us rely on the founders’ beliefs about whether AI is central to their firm, providing a broad measure of whether the firm is built around AI or not.<sup>2</sup> Especially given YC’s prominence and the reputational stakes involved,

<sup>1</sup>API available at <https://github.com/yc-oss/api>.

<sup>2</sup>The idea of building firms around AI could allow smaller firms to scale predates the launch of ChatGPT in late 2022. In 2017, five years before ChatGPT, a cofounder of InsureTech startup Lemonade urged founders to “don’t

there are strong incentives for these public statements to accurately reflect the firm’s underlying use of the technology.<sup>3</sup> Of the 2,891 YC startups in our sample, 990 (34.2%) are tagged as AI startups and 1,901 (65.8%) as non-AI startups (see Figure 1 for the distribution across batches in the PitchBook-matched regression sample and Appendix Figures A1–A2 for example YC pages illustrating this tagging).

### 3.1.2 Linking to PitchBook

We merge our YC data with PitchBook, a private-market data intelligence platform, using normalized company website URLs. PitchBook provides comprehensive data on startup financing, including deal-level funding amounts, valuations, and deal types.<sup>4</sup> Of our 2,891 YC firms, we successfully match 2,786 (96.4%) to PitchBook records. The small number of unmatched firms are primarily very early-stage companies without substantial financing activity or companies that have since become inactive. Appendix Table A3 consolidates the match rates for both the YC and PitchBook sample construction.

### 3.1.3 Linking to Revelio Labs

To measure organizational structure and firm size, we link our YC-PitchBook data to Revelio Labs, a workforce intelligence platform that aggregates information from public professional profiles and job postings to create comprehensive employee records. We use LinkedIn profile URLs from PitchBook to match companies to a pull of Revelio’s database covering records through January 2025, successfully linking 2,233 firms (77.2% of the full YC sample). Match rates are nearly identical for AI and non-AI startups (76.2% vs. 77.8%) and range from 69% to 82% across batches. Given that these are knowledge-work startups whose employees overwhelmingly maintain LinkedIn profiles, Revelio’s inability to link any employees to a firm strongly suggests the firm has effectively zero employees as of January 2025, so we impute zero employees for firms with no Revelio-matched workers. As our intensive margin results show, our findings are robust to coding non-matched

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hire more people, build smarter AI” and described his own firm as one designed around “fewer people and more code” (Wininger, 2017). As we discuss below, Lemonade’s vision mirrors our product channel. For example, embedding an ML system for predicting fraud into the product replaces the large teams that would otherwise manually review cases. Moreover, while ChatGPT led to a surge of AI startups in 2023, the first AI coding assistant is GitHub’s Copilot, which was launched in 2021. OpenAI’s GPT-3 was launched in 2020.

<sup>3</sup>Of course, self-disclosure risks strategic manipulation, similar to firms adding “.com” to their names during the internet bubble (Cooper, Dimitrov, and Rau, 2001). However, if the AI label is only being used superficially and symbolically, then while we might see exuberant or irrational shifts in valuation it seems unlikely that it would result in changes in how the firm is organized as measured by a third party data source (LinkedIn).

<sup>4</sup>That said, valuation and funding amounts are often missing for early stage firms like those we analyze.

startups as having missing values instead. Revelio provides position-level data including job titles, seniority levels, start and end dates, and functional classifications. Appendix Table A3 reports per-batch and field-level coverage rates.

## 3.2 PitchBook Sample: Extending to the Broader Startup Ecosystem

While YC provides a measurement of a startup’s AI focus and Revelio rich organizational data, it represents a highly selected sample of startups. To test whether patterns observed in YC generalize to the broader startup ecosystem, we extend our analysis to the universe of early-stage venture-backed startups using the PitchBook data.

### 3.2.1 Sample Construction

Appendix Table A3 (Panel B) reports the PitchBook side of our sample-construction funnel. We begin with all U.S.-based startups in PitchBook with a first financing deal between 2020 and 2024 (inclusive). The 2020–2024 first-deal window aligns with the YC W20–F24 sample period and the January 2025 Revelio employment snapshot, focusing on the post-GPT-3 era and ensuring every firm in the sample had at least one financing event before our outcome is measured. Using first deal date (rather than founding year) allows us to construct deal quarter fixed effects comparable to YC’s quarterly batch indicator.

### 3.2.2 Predicting AI Status

Unlike YC, PitchBook does not provide a clean indicator of whether a startup team wants to publicly disclose if it is an AI startup.<sup>5</sup> To classify startups as AI or non-AI, we train a machine learning model on the YC sample and then use the model to predict AI status for PitchBook companies using text descriptions of each startup.

In particular, we use PitchBook’s text descriptions of each company to generate 3,072-dimensional embeddings using OpenAI’s text-embedding-3-large model. We then train a Random Forest classifier to predict YC’s AI tags from these embeddings. We use 5-fold cross-validation to tune hyperparameters. The model achieves solid if unspectacular out-of-sample performance: cross-validated AUC of 0.81, and on a held-out test set, accuracy of 77.5%, precision of 73.9% (when we predict  $AI > 0.5$ , 74% are actually AI), and recall of 55.2% (we identify 55% of true AI companies). In

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<sup>5</sup>PitchBook assigns companies to “verticals” including an “Artificial Intelligence & Machine Learning” category, but this classification appears noisy, substantially undercounting AI startups relative to our YC-based measure. It is also unclear who applies this measure to the startup and what the incentives are for truthful reporting.

the paper’s regressions, we use the continuous predicted probability rather than a binary threshold as it provides more information. For descriptive statistics splitting the sample into AI and non-AI groups, we use a 0.5 threshold.

### 3.3 Variable Definitions

**AI Status.** For the YC sample, AI status is a binary indicator equal to 1 if the startup’s YC tags include “artificial-intelligence” or “ai.” For the PitchBook sample, AI status is the predicted probability from our Random Forest classifier trained on YC data. Since both variables range from zero to one, it is straightforward to compare effect sizes across the two.

**Employee Counts.** We measure employee counts using Revelio Labs data as of January 2025, which captures employees matched through public professional profiles. While our primary analyses use the January 2025 measures, we also construct a quarterly panel of employment counts to shed light on how firm size changes over time for AI and non-AI tagged firms.

**Manager Share.** The share of employees with O\*NET occupation codes beginning with “11-” (Management Occupations), based on Revelio’s mapping of job titles to standardized occupational categories.

**Hierarchy Depth.** The count of distinct Revelio seniority levels (1–7) present among a firm’s employees. To calculate a meaningful hierarchy measure, we require at least one employee in Revelio data, yielding a sample size of 2,233.

**Funding and Valuation Per Employee.** Total funding raised divided by employee count (funding per employee) and last known valuation divided by employee count (valuation per employee). As mentioned, funding and valuation amounts are often missing. For per-employee financing analyses, this yields 2,218 YC firms and 29,336 PitchBook firms with funding data, and 1,046 YC firms and 12,891 PitchBook firms with valuation data.

**Seniority Shares.** The share of employees at each seniority level based on Revelio Labs’ 1–7 seniority scale: Entry-level (1–2), Mid-level (3–5), and Senior (6–7). To calculate meaningful seniority share measures, we require at least one employee in Revelio data, again yielding a sample size of 2,233 firms.

**Functional Shares.** The share of employees in each functional category based on Revelio Labs job classifications: Engineering, Science, Sales, Marketing, Operations, Finance, and Administrative.

**Industry Group.** PitchBook assigns each company to one of 41 industry groups based on its primary business activity; examples include Application Software, Information Services, Healthcare Services, and Financial Software. We use PitchBook’s industry classification rather than standard SIC codes because SIC coverage is extremely sparse for early-stage startups. In our YC sample, fewer than 1% of firms (24 of 2,786) have SIC codes assigned in PitchBook’s data; in our broader PitchBook sample, fewer than 2% have SIC codes assigned. This low coverage likely reflects that SIC assignments in commercial databases are concentrated among more established firms.

**Batch.** For YC startups, the cohort identifier (e.g., W20, S21, W24) indicating when the startup was accepted into Y Combinator. Batches occur twice per year (Winter and Summer) plus occasional Fall batches.

**First Deal Quarter.** For PitchBook startups, the year-quarter of the company’s first recorded financing deal (e.g., 2020-Q1, 2023-Q3). This serves as a timing variable similar to YC batch.

### 3.4 Research Design

Our research design is descriptive, documenting if AI startups differ from non-AI startups in how they are organized, a question of significant scholarly and practitioner debate. Our approach follows in the spirit of recent work establishing stylized facts about who starts successful companies (Azoulay et al., 2020) and how AI tools are actually used across the workforce (Humlum and Vestergaard, 2025). Here, we provide the first systematic evidence on how firms built around AI are organized.

That said, two confounds are crucial to address to allow for meaningful comparisons between AI and non-AI startups. First, there is no discrete “arrival” of AI—the term dates to the 1950s. But since the launch of ChatGPT in late 2022, interest in AI tools and startups has exploded. As a result, there is the potential for strong cohort confounds: any observed difference between AI and non-AI firms might reflect *when* startups were founded rather than their AI status. A startup from the Winter 24 YC batch is both more likely to be AI and younger when we measure it in January 2025. We address this by including fixed effects for timing. In the case of our YC sample, we use the startup’s batch, effectively controlling for its quarter of entry. In PitchBook, we don’t have batch, and instead use first deal quarter.

Second, AI startups are likely to concentrate in different industries, so observed organizational differences could reflect industry differences and market structure rather than AI *per se*. Enterprise B2B software firms and consumer hardware firms naturally require different team structures regardless of whether they use AI. Yet, building on Chandler (1977), if building around AI causes firms to operate in different industries, and those industries then impact the nature of the organization, we could view this selection as an effect of AI. The railroad and telegraph similarly enabled firms to enter new industries and serve national markets, which in turn shifted their organizational structure. Here, we take the more conservative approach and focus on within-industry comparisons to better isolate organizational differences from industry differences.

Our baseline specification for the YC sample is:

$$Y_i = \beta \cdot \mathbf{1}[\text{AI}_i] + \gamma_{bg} + \varepsilon_i \quad (1)$$

where  $Y_i$  is a firm-level measure for startup  $i$ ,  $\mathbf{1}[\text{AI}_i]$  indicates whether the startup is classified as AI, and  $\gamma_{bg}$  are batch  $\times$  industry fixed effects. For the PitchBook sample, we replace the binary AI indicator with the continuous AI probability score and replace batch with first deal quarter, using first deal quarter  $\times$  industry fixed effects.

By including these interacted fixed effects, we compare AI and non-AI startups operating in the same industry that entered in the same cohort, ensuring our AI coefficient captures within-industry-cohort variation rather than cross-industry differences or simple cohort timing confounds. We report heteroskedasticity-robust (HC1) standard errors throughout.

## 4 Results

### 4.1 Summary Statistics and Growth of AI Startups

We begin by documenting basic patterns in our data. Table 1 presents summary statistics for our two samples. Even before conditioning on batch or industry, we observe raw differences. AI startups are substantially smaller than non-AI startups: the mean AI startup in our YC sample has approximately 13 Revelio-matched employees compared to 42 for non-AI startups (based on January 2025 snapshots)—a difference of nearly 70%. AI startups also have flatter hierarchies (3.8 vs. 5.0 seniority levels), fewer managers (17% vs. 24%), more engineers (45% vs. 36%), and fewer entry-level workers (19% vs. 30%). Despite their smaller size, AI startups have comparable total



funding and valuations to non-AI firms, implying that AI startups are substantially more capital-efficient on a per-employee basis. That said, caution must be taken when comparing these means, as they fail to control for cohort or industry composition differences. The summary statistics for the PitchBook sample (Panel B) show similar patterns, though more muted—likely because PitchBook includes a much broader sample of firms across more industries. This suggests that within-industry comparisons may be necessary to reveal differences in the broader sample.

Indeed, when we turn to Figure 1 we can see the dramatic rise of AI startups in both samples. The share of YC startups self-identifying as building around AI has increased substantially, rising from approximately 15% in early 2020 batches to over 50% in 2024 batches. This pattern is mirrored in our broader PitchBook sample, where the share of early-stage startups we classify as AI has grown from roughly 8% in 2020 to over 20% by 2024. This also underscores the need to compare within cohorts, as AI startups in our data are from much more recent vintages.

Figure 2 shows the share of AI and non-AI startups across YC sub-verticals (YC’s own finer-grained and startup-focused industry classification), ordered by relative AI concentration. AI startups cluster in sub-verticals like legal services and developer tools, where applying AI is thought to have especially large impacts on worker productivity (Eloundou et al., 2024a). Non-AI startups are more prevalent in hardware-intensive sub-verticals like aviation and medical devices.

## 4.2 How Are AI Startups Organized?

We first examine whether AI startups have smaller teams. We start with the raw data. Figure 3 shows a histogram of employee counts, confirming that the modal AI startup has fewer than 10 employees, whereas non-AI startups are more evenly distributed across size categories. Given that there has been an increase in the share of AI startups, these differences could simply be due to cohort effects. However, in Figure 4 we plot the size of AI and non-AI firms relative to the number of quarters since they graduated from YC. AI firms are roughly half the size of non-AI ventures and this gap persists for at least three years. Appendix Figure A3 presents the raw distributions separately for each YC batch, showing that the distributional difference in Figure 3 between AI and non-AI startups are evident within each cohort. Finally, these results are not rooted in industry differences either: Appendix Figure A4 presents kernel density plots of the firm size distribution for AI and non-AI startups, with batch-industry fixed effects residualized out. The distributions are visibly shifted, with AI startups concentrated at smaller sizes.

Table 2 reports regression estimates for the relationship between AI status and firm size, com-

paring startups within the same cohort and industry. In levels, AI startups in the YC sample have approximately 8 fewer employees than non-AI startups, compared to a non-AI mean of 34 employees. In the PitchBook sample, AI startups have approximately 6 fewer employees off a non-AI mean of 18 employees. The log specifications, which examine the intensive margin conditional on having at least one employee, show that AI startups are roughly 25% smaller in the YC sample (coefficient of  $-0.28$ ) and 12% smaller in the PitchBook sample (coefficient of  $-0.13$ ).<sup>6</sup> Results are robust across specifications: levels, logs, and Poisson models all show significant negative effects of AI status on firm size. Appendix Table A4 shows these patterns hold when we use alternative measures of firm size sourced from PitchBook and YC rather than Revelio.

These teams are not just smaller but employ a different mix of functional expertise. Table 3 examines the composition of startup workforces across seven functional categories. AI startups employ a significantly higher share of technical workers relative to business and administrative staff. The coefficient on AI status when predicting engineering share is positive and significant in both samples ( $\beta = 0.045$ ,  $p < 0.01$  in YC;  $\beta = 0.183$ ,  $p < 0.01$  in PitchBook), indicating that AI startups allocate roughly 5 percentage points more of their workforce to engineering roles in YC and 18 percentage points more in PitchBook. Conversely, AI startups have significantly lower shares of employees across business functions: sales ( $\beta = -0.021$ ,  $p < 0.05$ ), operations ( $\beta = -0.009$ ,  $p < 0.05$ ), finance ( $\beta = -0.017$ ,  $p < 0.01$ ), and administrative roles ( $\beta = -0.011$ ,  $p < 0.05$ ). The coefficient on marketing share is negative but not statistically significant in the YC sample ( $\beta = -0.012$ ). The statistically significant declines span both external-facing functions (sales) and internal process functions (finance, operations, administration) in YC, and extend to all business functions including marketing in the broader PitchBook sample. This suggests that the talent needed to scale firms may shift when they are built around AI (Bresnahan, Brynjolfsson, and Hitt, 2002; DeSantola, Gulati, and Zhelyazkov, 2022; Lee and Kim, 2024). These patterns are even more pronounced in the broader PitchBook sample, where all functional coefficients are significant at the 1% level. We interpret these cautiously, however, because AI status in this sample is inferred from a text-based prediction, which may mechanically inflate compositional differences. The picture that emerges is that AI startups are technical-heavy and business-function-light—consistent with shifting the firm’s production function toward the technical talent needed to build and maintain AI

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<sup>6</sup>For small coefficients,  $e^\beta - 1 \approx \beta$ , so log coefficients can be read directly as approximate percentage changes. For larger coefficients, the exponential transformation is more accurate. Throughout the paper, we convert log coefficients to percentage changes using the transformation  $e^\beta - 1$ , which provides an exact interpretation for discrete changes in AI status.

capabilities. That said, it is important to remember that these firms are smaller, thus while there are relatively more engineers, there are fewer employees in each role across the board for AI-native firms.

We next turn to workforce seniority. Table 4 examines the seniority composition of AI versus non-AI startups. AI startups have 5 percentage points more senior employees ( $\beta = 0.052, p < 0.01$ ) and 4 percentage points fewer entry-level employees ( $\beta = -0.040, p < 0.01$ ). The broader PitchBook sample shows the same pattern, with PB AI probability associated with a 6 percentage point lower entry-level share ( $\beta = -0.060, p < 0.01$ ) and a 6 percentage point higher senior share ( $\beta = 0.056, p < 0.01$ ). This is consistent with Brynjolfsson, Chandar, and Chen (2025), who document a 13% relative decline in early-career employment in AI-exposed occupations. At the task level, a growing body of evidence suggests that AI often disproportionately benefits less experienced workers (Brynjolfsson, Li, and Raymond, 2023; Dell’Acqua et al., 2023; Noy and Zhang, 2023; Csaszar, Ketkar, and Kim, 2024). One might therefore expect AI-native firms to employ more junior talent. We find the opposite: AI-native firms employ fewer entry-level workers and more senior workers. This contrast underscores the importance of studying AI at the organizational level. Who benefits from AI depends not only on task-level productivity gains, but also on where AI is applied in the firm’s production process (Gans and Goldfarb, 2026; Agrawal, Gans, and Goldfarb, 2024; Benzell and Myers, 2026; Otis et al., 2025; Kang and Kim, 2025).

Given that the teams are smaller, more senior, and more technical, we also examine whether hierarchy differs. Figure 5 shows the distribution of hierarchy depth by AI status. AI startups have significantly flatter organizational structures: the median AI startup has 3 distinct seniority levels compared to 4 for non-AI startups. Table 5 confirms this pattern in regression estimates, with AI startups having roughly 0.5 fewer hierarchy levels on average ( $\beta = -0.482, p < 0.01$ ). In Appendix Table A5 we show these findings are robust to controlling for firm size—the flattening is not simply a mechanical consequence of being smaller (see also Appendix Figure A5 for the residualized distribution).

Consistent with less need for coordination in AI startups, these firms also have a lower share of employees in management occupations. Using O\*NET occupation codes to identify managers (codes beginning with “11-”), we find that AI startups have approximately 4 percentage points fewer managers as a share of total employment ( $\beta = -0.037, p < 0.01$ ). These results are reminiscent of the flattening of firms due to information technology (Brynjolfsson et al., 1994; Bresnahan, Brynjolfsson, and Hitt, 2002; Rajan and Wulf, 2006; Bloom et al., 2014), suggesting that when

productive capability moves from people to the product, the need to coordinate and supervise internal knowledge work may diminish.

### 4.3 Do AI Startups Achieve More with Less?

The preceding findings raise a natural concern: perhaps AI startups are smaller and flatter simply because they are lower quality or have less potential. AI may lower entry barriers, allowing anyone to enter, even with a weak idea (Reimers and Waldfogel, 2026). If so, the organizational differences we document would reflect the limitations of building around AI rather than evidence that AI is changing the nature of the firm.

We begin by examining total funding and valuation in levels. Table 6 (Panel A) reports regression estimates. In the YC sample, AI startups raise approximately 15% *less* total funding ( $\beta = -0.165$ ,  $p < 0.05$ ) and show no statistically significant difference in valuations. In the broader PitchBook sample, the pattern reverses: AI startups raise approximately 15% more funding ( $\beta = 0.136$ ,  $p < 0.10$ ) and achieve 58% higher valuations ( $\beta = 0.458$ ,  $p < 0.01$ ). This divergence likely reflects YC’s selection: virtually all YC startups are technology firms that have met a high admission bar, compressing differences in funding and valuation relative to the broader PitchBook universe.

The more informative comparison is value per employee. In Figure 6, the raw distributions of valuation per employee show that AI startups are shifted rightward, achieving higher valuations relative to their headcount. Appendix Figure A6 residualizes out batch fixed effects and shows a similar pattern.<sup>7</sup>

Table 6 (Panel B) quantifies these patterns, focusing on the subset of startups with available funding and valuation data in PitchBook.<sup>8</sup> In the YC sample, AI startups raise approximately 12% more funding per employee ( $\beta = 0.109$ ,  $p < 0.10$ ) and result in 30% higher valuation per employee ( $\beta = 0.260$ ,  $p < 0.05$ ). The PitchBook sample confirms these patterns more strongly: AI startups raise 33% more per employee ( $\beta = 0.285$ ,  $p < 0.01$ ) and achieve 76% higher valuation per employee ( $\beta = 0.564$ ,  $p < 0.01$ ). These efficiency results are consistent across both samples and robust to specification.

Notably, the valuation-per-employee effects are substantially larger than the funding-per-employee

<sup>7</sup>Valuations reflect investors’ expectations of future cash flows rather than realized value. We use valuation per employee as a measure of capital-market-implied productivity, not as a direct measure of output.

<sup>8</sup>Funding-per-employee results are robust to treating missing funding values as zeros, which has a defensible interpretation (no recorded deal). For valuation, missing values reflect “no priced round on record” rather than literal zero value (no firm in our sample has a PitchBook-recorded valuation of exactly zero), so we restrict the valuation sample to firms with a non-missing valuation rather than imputing zeros. Poisson estimates yield similar conclusions for funding outcomes.

effects in both samples. If AI startups were simply raising more capital, we would expect funding and valuation to increase proportionally. Instead, the disproportionate increase in valuation relative to funding is consistent with investors expecting higher future productivity from AI startups conditional on the capital and labor inputs.

#### 4.4 Where Do AI Startups Locate and Who Works for Them?

A large literature demonstrates that as the nature of organizations changes, so too do the locations where firms concentrate and the workers they employ (Botelho, Gulati, and Sorenson, 2024). New technologies can have disproportionate impacts on particular groups—shaping employment opportunities and potentially long-term career prospects (Feigenbaum and Gross, 2022). We briefly examine these patterns for AI startups, reporting figures and tables in the Appendix.

Appendix Table A8 shows that AI startups are more geographically concentrated than non-AI startups. AI startups are 20 percentage points more likely to be headquartered in San Francisco ( $p < 0.001$ ), a substantial increase relative to the baseline rate of just under a quarter among non-AI startups. This concentration likely reflects the importance of access to specialized AI talent and proximity to leading AI research labs, consistent with longstanding findings on the distinctive advantages of Silicon Valley as a technology hub (Guzman and Stern, 2015; Saxenian, 1994). These geographic patterns hold in the broader PitchBook sample as well (Appendix Table A11).

We also find demographic and educational differences between AI and non-AI startup workforces (Appendix Tables A6–A10). These demographic patterns are consistent with the expert-dense organizational form we document: the shift toward technical, senior workforces draws from labor pools that remain less diverse. In the YC sample, AI startup employees are about 5 percentage points more likely to hold graduate degrees (Appendix Table A7). In the broader PitchBook sample, AI probability is associated with a 5.9 percentage point lower female share, an 8.5 percentage point higher Asian/Pacific Islander share, a 17.6 percentage point higher graduate-degree share, and higher university prestige (Appendix Tables A9 and A10).<sup>9</sup>

These geographic and demographic patterns raise questions about whether the benefits of AI-enabled organizational forms will accrue evenly, given persistent disparities in entrepreneurial finance by gender and race (Ewens, 2022; Marx, Wang, and Yimfor, 2025). Evidence suggests that women

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<sup>9</sup>Revelio Labs computes prestige scores using an expectation-maximization model that alternates between user and institution prestige updates, with world university rankings as priors. Prestige values are continuous between  $-1$  and  $1$ , weighted by seniority and tenure. See <https://www.data-dictionary.reveliolabs.com/methodology.html> for full methodology.

adopt generative AI at lower rates than men across multiple contexts, even after controlling for occupation, education, and technical skill (Otis et al., 2026). If AI tools accelerate learning for those who use them, differential adoption rates may translate into widening performance gaps—both for individual workers within firms and for the entrepreneurs who found them.

## 5 Exploring AI Product and AI Process

The results so far establish that AI-native firms are smaller, flatter, more technically concentrated, and more capital-efficient than non-AI startups in the same industry and cohort. But these patterns alone do not tell us *why* AI startups are organized differently. As discussed in Section 2, we note that two primary channels could account for these patterns: workers using AI tools to do their jobs more productively (the process channel), and AI capabilities embedded directly into the firm’s product (the product channel). In this section we measure each channel separately and explore how each accounts for the organizational differences we documented above.

### 5.1 Measuring AI in a Firm’s Product and Processes

To quantitatively measure the product channel, we begin by classifying how founders describe the role of AI in their businesses. To do so, we built an inductive classification of the descriptions founders post of their startup on YC’s startup directory platform. To generate the classification, we first drew a random sample of 40 AI-tagged startup descriptions and read them qualitatively, identifying recurring patterns in how AI enters the product. We formalized those patterns into four categories and then used an LLM to apply the schema to the 990 AI-tagged YC startups (see Appendix for the full methodology and prompt). Figure 7 presents the resulting distribution of AI-tagged startups, and Table 7 provides concrete examples from each category. For the regression-based AI Product measure we use below, we re-apply the same schema to the full YC sample so AI and non-AI firms are coded symmetrically.

The classification captures different ways that startups embed AI into the product. The largest category, *autonomous workflow* products (43%), includes startups whose products use AI to perform tasks or roles that previously required human labor. Artisan (W24) builds AI sales representatives that discover leads, research prospects, and write personalized emails. Aurelian (S22) deploys AI phone operators that triage 911 calls. These products place AI directly in the workflow, allowing the product itself to perform work that would otherwise require salespeople or operators. The startup

can offer the service without building a large internal organization to deliver the underlying work itself.

The second category, *expert augmentation* products (24%), includes startups whose products use AI to extend a professional’s existing workflow. Tower (W24) helps lawyers stay organized and close deals faster. Medium Biosciences (S21) uses AI to design therapeutic proteins, augmenting biologists in drug development. Here, AI does not replace the expert role outright. Instead, the product embeds analytic, generative, or search capabilities that expand what expert users can do. This also moves productive capability into the product: rather than hiring large internal teams of lawyers or scientists to provide expert services directly, the startup builds a product that gives users some of those capabilities themselves.

These first two categories are clear examples of startups embedding AI into their products. Whether the startup is augmenting or automating work, in both cases the company is building a product around AI. Taken together, we find that about 67% of YC startups tagged as AI-native are clearly embedding AI directly into the products they sell.

The third category, *AI infrastructure* products (15%), includes firms whose products help others build, deploy, or use AI. Exa (S21) offers an AI-powered search engine using embeddings and neural retrieval, and Roboflow (S20) provides a computer vision platform for deploying object detection models. These firms embed AI into tools that become part of other companies’ production systems. The product does not always automate an end-user workflow directly, as in autonomous workflow products, or augment a specific professional role, as in expert augmentation products. Instead, some packages model capabilities, data pipelines, or deployment tools into infrastructure that other firms can build on. Thus, some of these firms clearly embed AI in the products they sell while others sell infrastructure that complements the use of AI (e.g., monitoring and analytics tools).

Finally, the remaining 18% (“Other”) includes startups where AI’s role is present but harder to classify, including games, robotics platforms, or firms whose descriptions do not clearly specify how AI is embedded in the product.

To create a proxy for the process channel, we operationalize internal AI tool use directly from firms’ job posting text. For each firm in our sample we collect all Revelio Labs postings dated through January 2025—the same window over which we measure employment—and drop scrapes flagged as staffing-firm aggregator listings, instead focusing on postings directly listed by the hiring firm. We then code a firm-level binary indicator, AI Process, equal to one if any of the firm’s non-staffing postings names a specific Gen AI assistance tool from a dictionary of 33 AI tools. The

dictionary includes general-purpose AI assistants (e.g., ChatGPT, Claude), coding agents (e.g., GitHub Copilot, Cursor), creative tools (e.g., Midjourney, Adobe Firefly) and workflow products (e.g., Granola, Mem.ai). The dictionary was assembled by reading the corpus of job postings and noting listed AI tools along with tools included in the Stack Overflow 2024 Developer Survey of AI-tools section. The full dictionary and construction details are listed in Appendix A.3.

Two example postings from AI-tagged YC firms illustrate why focusing on tools, rather than generic mentions of “AI,” is more likely to capture process-focused uses. The first, from Overlap (S24), a multimodal-AI firm hiring a Product Engineer, is saturated with AI language—“multimodal AI agents,” “Video Marketing AI,” “AI-powered products,” “AI-driven media”—but every reference describes the AI features that will be built into the product. No specific AI tool, model, or assistant from our dictionary is named, so AI Process equals zero. The second, from Fintool (W23) hiring a Full-Stack Engineer, contains a clean process-channel signal: “We are heavy users of V0, Cursor and other AI tools.” That sentence describes the firm’s *workers* using specific named AI tools to ship product features, and it matches the V0 and Cursor patterns in our dictionary, so AI Process equals one. The channel regressions use the 1,684 firms with at least one non-staffing posting through January 2025; the log, hierarchy, manager-share, and base-rate analyses further require at least one Revelio-matched employee, leaving 1,666 firms (467 AI-tagged and 1,199 non-AI startups).

Within this sample of 1,666 firms with at least one job posting, the product channel is present at 69.8% of AI-tagged firms versus 16.1% of non-AI firms, while the process channel is present at 13.5% of AI-tagged firms versus 5.2% of non-AI firms—a roughly  $2.6\times$  relative gap on the process side, though with substantially smaller absolute levels than the product channel. These lower levels likely reflect both the nascency of AI assistance tooling over our sample window and the fact that job postings likely underreport workers’ use of AI tools at startups. Still, the pronounced difference between AI and non-AI firms provides further evidence that YC’s AI tag does not merely reflect strategic disclosure, but captures real differences in what and how the startups are being built.

To shed light on how AI shapes how firms are organized, Table 8 regresses Revelio employment and hierarchy outcomes on our measures of AI product and process channels. For the product channel, we use a dummy variable capturing if the startup was classified as embedding AI into its product. For the process channel, we use a binary measure of whether any job posting matches our list of AI tools, a noisier proxy. Every column controls for log non-staffing postings and absorbs batch  $\times$  industry fixed effects; the coefficient on the posting control is suppressed in the table. We find that the product channel is a key driver of the AI organizational pattern, controlling for



internal AI tool use: firms we classify as producing AI products have roughly 10 fewer employees than otherwise-comparable peers in the same batch and industry (column 1,  $\beta = -10.2$ ,  $p < 0.05$ ), are about 13% smaller on the intensive margin (column 2,  $\beta = -0.144$ ,  $p < 0.01$ ), with a similar 19% gap in Poisson (column 3,  $\beta = -0.212$ ,  $p < 0.01$ ), and operate with about 0.2 fewer hierarchy layers (column 4,  $\beta = -0.211$ ,  $p < 0.05$ ). By contrast, we find limited evidence for the process channel: its levels estimate is positive ( $\beta = 45.2$ ,  $p < 0.10$ ), while the log, Poisson, hierarchy, and manager-share estimates are not statistically significant.

## 5.2 When are AI-native firms most different?

When might the impacts of AI be largest? Building on the idea that embedding AI into a product enables knowledge work to scale with many fewer workers, we should expect the differences we observe to be largest for businesses that, before AI, scaled by hiring people to deliver the underlying work. In the context of YC startups, this points to services firms in areas like therapy, counseling, and tutoring, where delivery has historically required either an internal workforce or a marketplace of gig workers. When AI can do that work inside the product, the firm no longer needs to staff or coordinate that labor, potentially enabling radically smaller firms to scale.

To explore this channel, we started by identifying pairs of startups in our YC data that are near in product space, but vary dramatically in the number of workers they employ. To do this, we used OpenAI `text-embedding-3-large` representations of each startup’s YC description (3,072-dimensional vector embeddings) to find each AI-tagged YC startup’s nearest non-AI peers in product space using cosine similarity, searching within the same batch-industry cell when possible and otherwise within the same industry across batches. Reviewing the pairs, we found a number of examples consistent with the idea that embedding AI in the product enables “services” businesses to shift from workforces to compute. For example, *Educato AI*, an AI-native exam-prep platform, has 7 employees; *Careerist*, a non-AI platform that delivers courses through third-party tutors, has 912. Instead of people developing and communicating the content, the AI does, consistent with the idea that embedding AI into a product allows small teams to scale. *Legion Health*, an AI psychiatry platform built on LLMs and AI agents, has 28 employees; *Intellect*, a non-AI mental healthcare firm built on a network of human therapists, has 446. Again, we see a services model that scaled with human labor shift towards one built around AI that scales with compute. While these pairs are extreme and highly selected cases, they illustrate that AI’s impact on firm organization is likely to be largest when it shifts a startup offering a service from doing that work with people to doing it

with compute.

Finally, to test this idea more systematically, we group PitchBook industry groups into three buckets. *Software*, the largest single category, serves as the reference. *Services-delivery* captures industry groups whose pre-AI delivery model typically required scaling worker-hours with revenue: healthcare (four groups spanning clinical services, health-technology systems, and related care), commercial and non-financial services, and media. *Other* captures everything else, including pharma and biotech, hardware, financial technology, healthcare devices, and agriculture. Table 9 reports our headline outcomes interacted with these buckets.

The estimates are consistent with our case studies. In the Software reference category, AI startups are roughly 25% smaller than non-AI peers ( $\beta = -0.286$ ,  $p < 0.01$ ). The interaction for Other is effectively zero ( $\beta = -0.101$ , n.s.). The interaction with services-delivery adds 93 log points of shrinkage on top of the baseline coefficient ( $\beta = -0.929$ ,  $p < 0.01$ ), which implies that AI startups in services-delivery operate at roughly 30% the headcount of non-AI peers in the same batch and industry. Hierarchy flattens similarly: the baseline estimate is  $-0.375$  fewer layers ( $p < 0.01$ ), and services-delivery adds even more again ( $\beta = -0.450$ ,  $p < 0.05$ ). Taken together, these examples and estimates suggest that embedding AI into products is a crucial channel for understanding how AI will impact the organization of firms.

## 6 Discussion and Conclusion

Technology has long reshaped organizational form, and there is no shortage of speculation that AI will do the same. We find that AI-native startups are 25% smaller, half a seniority level flatter, and substantially more expert-dense than non-AI startups in the same industry and cohort, yet they carry comparable valuations. These organizational differences are concentrated among firms that embed AI in their products: about two-thirds of AI-native startups in our data build AI directly into what they sell, and the headcount and hierarchy gaps relative to non-AI peers are largest in industries where the pre-AI workforce delivered services to customers. Our findings on how AI startups are organized complement emerging evidence that AI is already reshaping firm entry and performance (Asam and Heller, 2025; Cao and Bhatia, 2025; Gupta et al., 2025).

Our study is not without limitations. Our measure of AI status relies on self-identification, which may miss firms building around AI without publicly claiming that identity or may include firms that strategically adopt the label. Our cross-sectional design cannot pin down causal mechanisms:

the patterns we observe could stem from the product architecture, from AI-enabled efficiencies in internal operations, or from selection of different types of founders into AI ventures. Our data covers firms for at most a few years; whether these organizational forms persist as firms scale, or whether the demands of coordination reassert themselves, is something we cannot yet address (Joseph and Sengul, 2025).

It is also crucial to note that our results do not cleanly map onto labor market outcomes. Each AI-native firm in our data is smaller than its non-AI peers, but aggregate AI-segment employment need not fall if AI also enables many more startups to enter (Bena, Bian, and Giannetti, 2026). In our YC data, AI-native firms employ about half as many workers as non-AI peers after three years (Figure 4), implying the entry margin would need to roughly double to offset the headcount gap. In PitchBook, AI first-deal counts in 2024 are nearly 8 times the 2020 average. While some of this entry is surely displacing non-AI ventures, the idea that AI enables enough new firms to lead to *more* labor demand is not out of the question. Mapping these broader employment dynamics is an important question for future work.

Our measures also offer a useful complement to existing work on AI’s labor market effects. Most existing research uses exposure measures to capture where AI could affect labor demand (Eloundou et al., 2024b; Felten, Raj, and Seamans, 2024; Bena, Bian, and Giannetti, 2026) and to estimate employment responses to generative AI (Gupta et al., 2025; Brynjolfsson, Chandar, and Chen, 2025). Exposure measures are valuable, but they tell us where AI *could* matter rather than where AI is actually being used, and they risk picking up macro trends correlated with exposure. Our paper complements this exposure work by directly measuring whether startups identify as AI-native, whether they have embedded AI in their product, and whether they are adopting AI tools in their processes.

Our paper makes an important conceptual contribution by distinguishing between AI’s impact on processes and products. Recent experimental work has focused primarily on the process channel, and experiments show that when firms learn to map AI into their production, AI can have meaningful effects on revenue growth and capital demand (Kim, Kim, and Koning, 2026). Here we show the importance of the product channel. Through both our case studies and our quantitative analyses, we offer an account of why AI-native firms look the way they do: these firms move much of what used to be internal knowledge work directly into the product, allowing smaller and flatter teams to scale.

More broadly, our findings suggest that AI may not simply make existing organizations more

efficient—it may change what organizations look like and do. If firms can increasingly import capabilities from foundation models rather than build them through people, they may no longer be constrained by human attention and processing and the managerial problem shifts from accumulating internal capacity to building and integrating external capabilities into products and production. The startups we study offer an early window into this shift. Understanding which firms solve this organizational problem, and how, is a central question for future research on how AI will change entrepreneurship and firm strategy.

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## 8 Figures

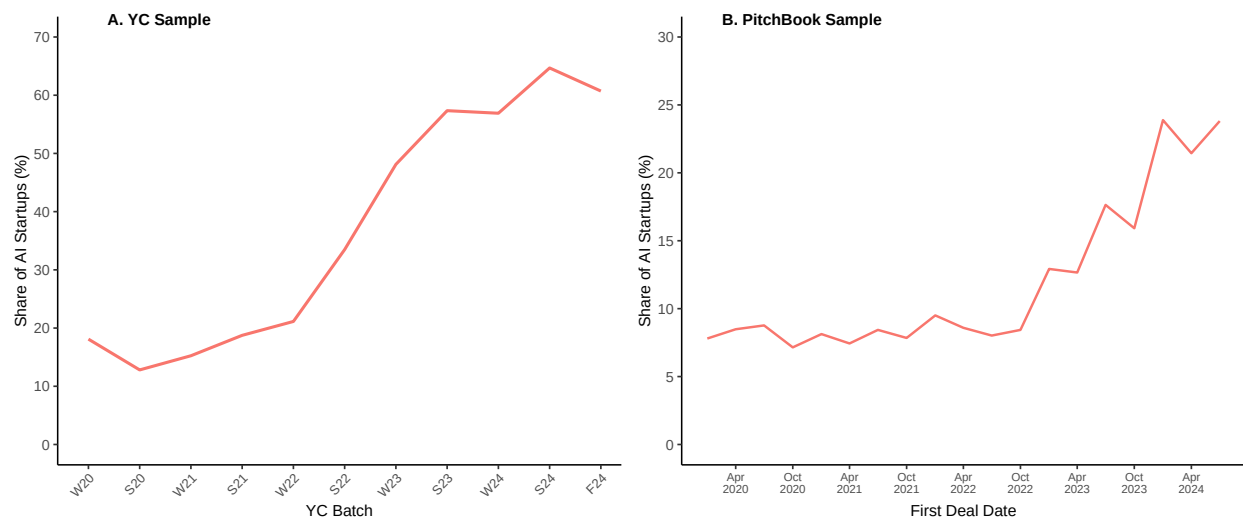


Figure 1: Share of AI Firms over Time. Panel A shows what share of Y Combinator startups self-identified as AI across batches W20 through F24, on a sample of 2,786 firms. Panel B shows what share of PitchBook startups our classifier labels as AI, by first deal quarter, between the start of 2020 and Q3 of 2024. The Panel B sample is 45,313 firms, of which 5,599 are classified as AI. We cap Panel B at 2024Q3 because deal coverage in PitchBook drops dramatically in our data. YC AI status is a binary tag chosen by the startup. PitchBook AI status is a Random Forest prediction trained on YC labels.

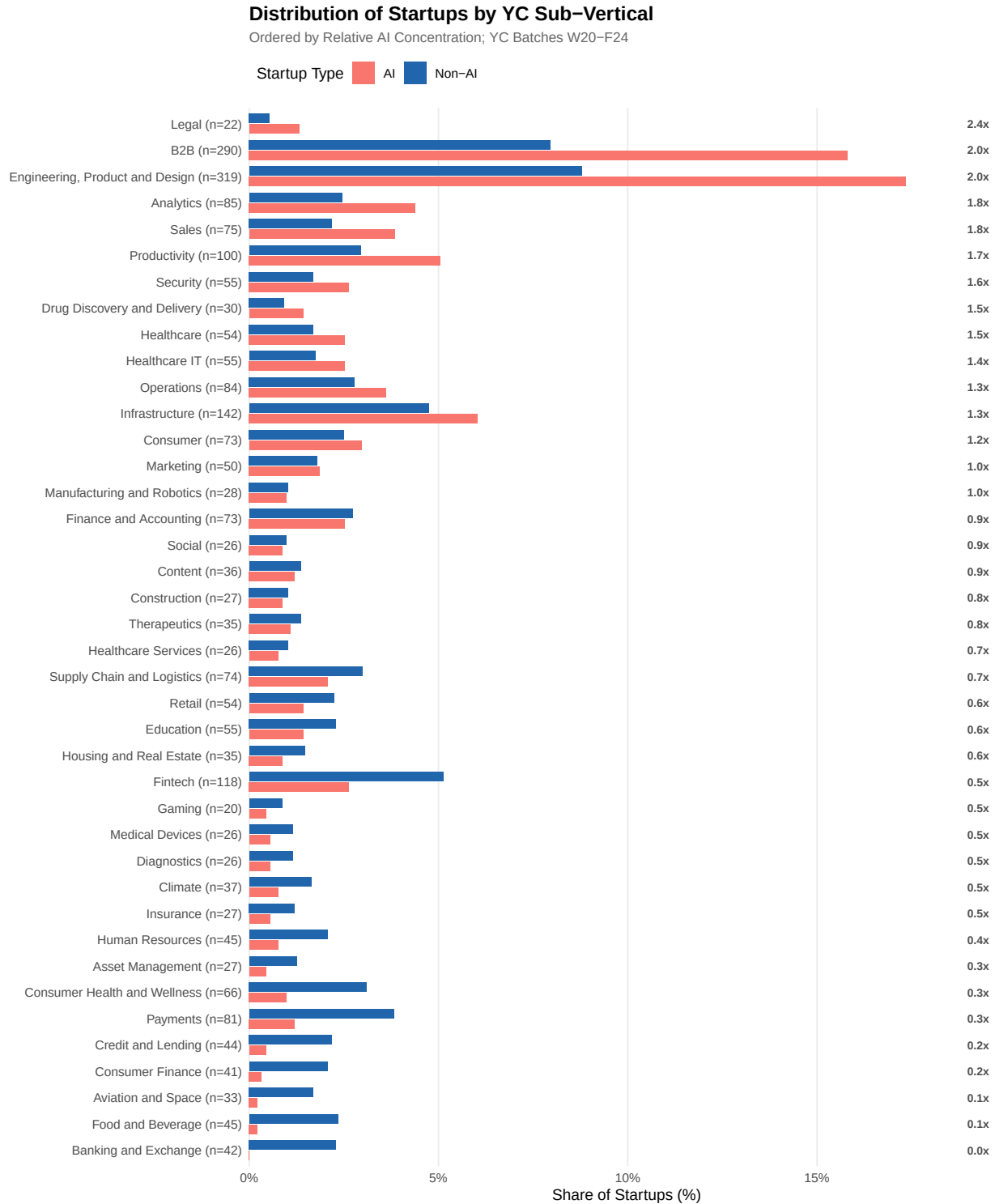


Figure 2: Distribution of Startups by YC Sub-Vertical. The figure shows the share of AI (red) and non-AI (blue) startups across the 40 YC sub-verticals with at least 20 firms (of 57 total, after excluding the “Unspecified” bucket), ordered by relative AI concentration. AI startups concentrate in knowledge-work sub-verticals; non-AI startups concentrate in healthcare, finance, and hardware sub-verticals.  $N = 2,786$ .

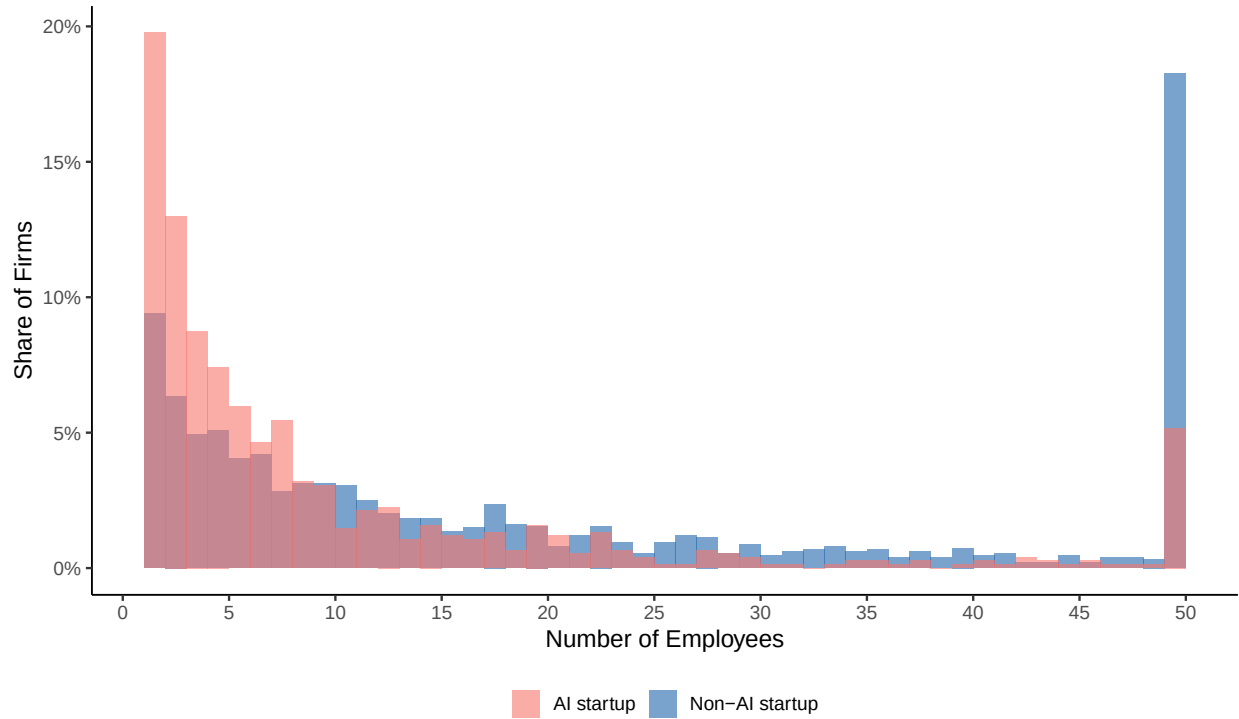


Figure 3: Histogram of Employee Counts by AI Status. Distribution of employee counts for AI (red) and non-AI (blue) YC startups. Sample is the workforce sample that includes all YC firms from batches W20–F24 with at least one employee in the Revelio data,  $N = 2,233$ . Values winsorized at 50 employees (approximately the 95th percentile). Employee data from Revelio Labs as of January 2025.

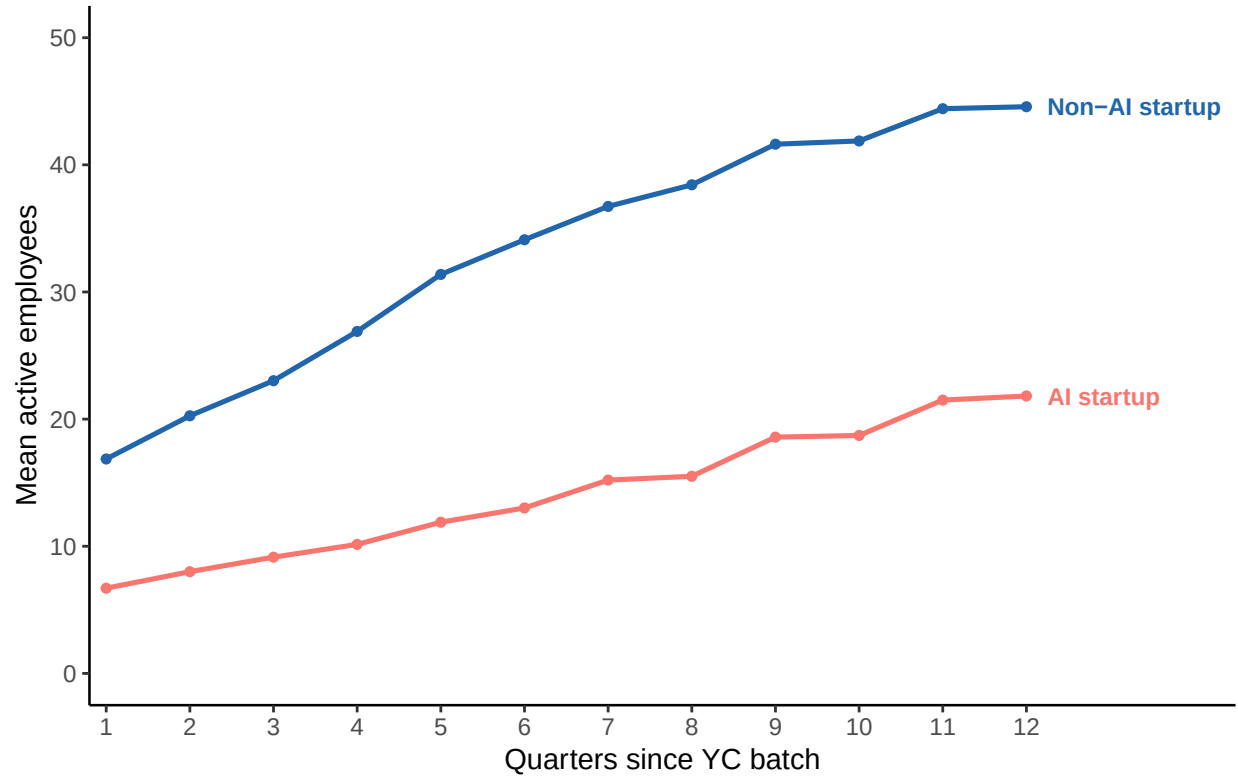


Figure 4: The figure plots mean active Revelio employees by quarters since YC batch for AI-native (red) and non-AI (blue) YC startups. Firms with no matched active Revelio employees in a quarter are coded as zero. The sample is W20–F24 firms with non-missing PitchBook IndustryGroup; the window starts one quarter after YC batch and covers up to the next three years.

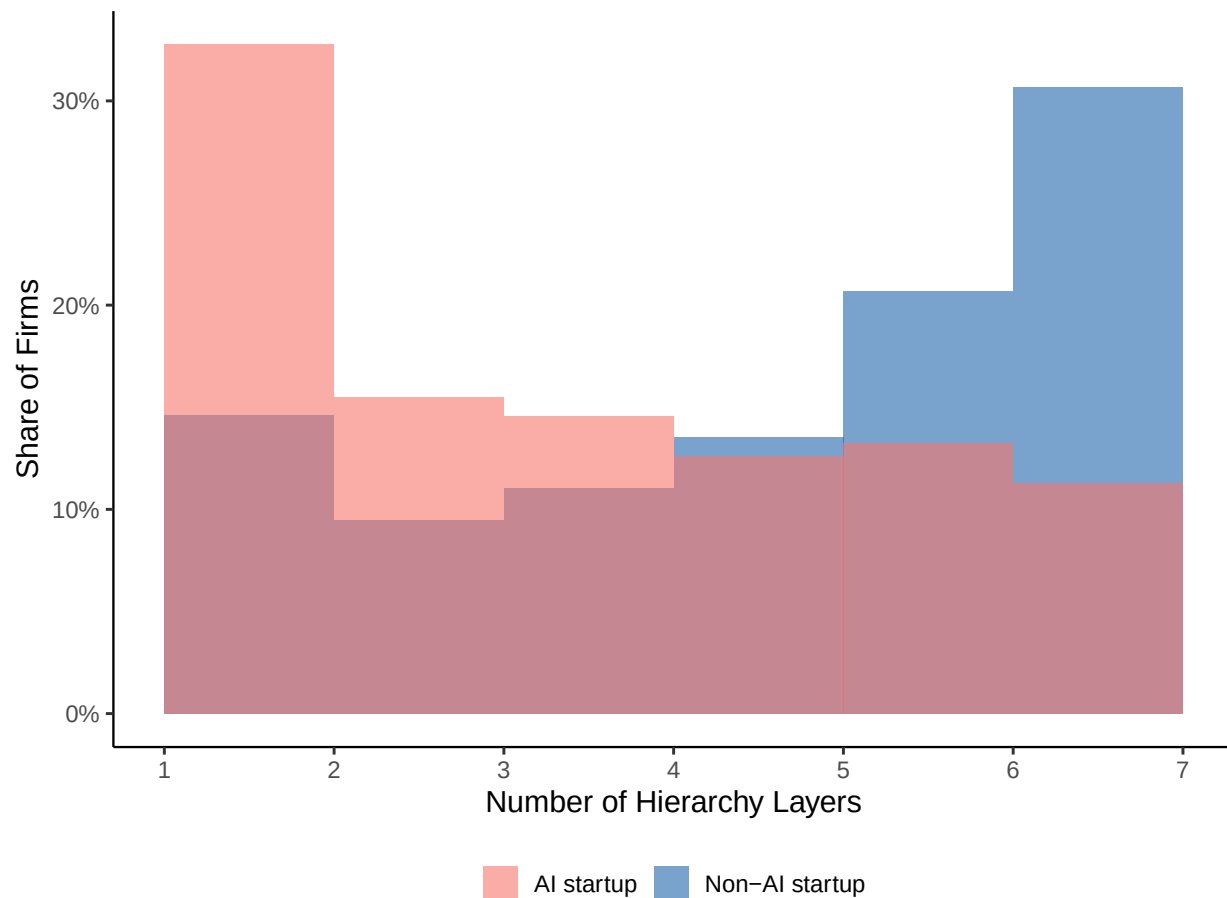


Figure 5: Distribution of Hierarchy Depth by AI Status. Distribution of organizational hierarchy depth for AI (red) and non-AI (blue) Y Combinator startups. Hierarchy depth is measured as the count of distinct Revelio seniority levels (1–7 scale) present among a firm’s employees. Sample is the workforce sample that includes all YC firms from batches W20–F24 with at least one employee in the Revelio data,  $N = 2,233$ . Employee data from Revelio Labs as of January 2025.

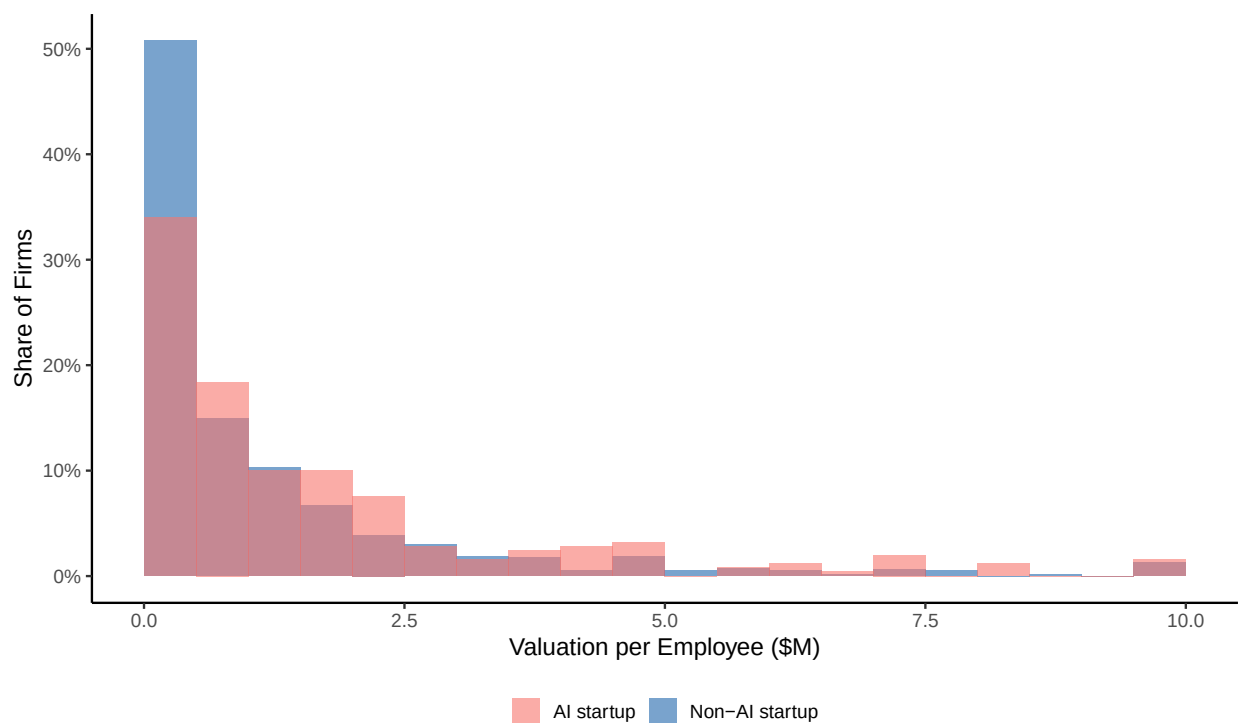


Figure 6: Histogram of Valuation per Employee by AI Status. Distribution of valuation per employee for AI (red) and non-AI (blue) Y Combinator startups from batches W20–F24 with both a Revelio-matched employee count and a non-missing PitchBook valuation,  $N = 1,046$ . Values winsorized at \$10 million per employee (approximately the 97th percentile). Valuation per employee is calculated as last known valuation from PitchBook divided by employee count from Revelio Labs as of January 2025. Firms with no recorded valuation in PitchBook are excluded rather than coded as zero.

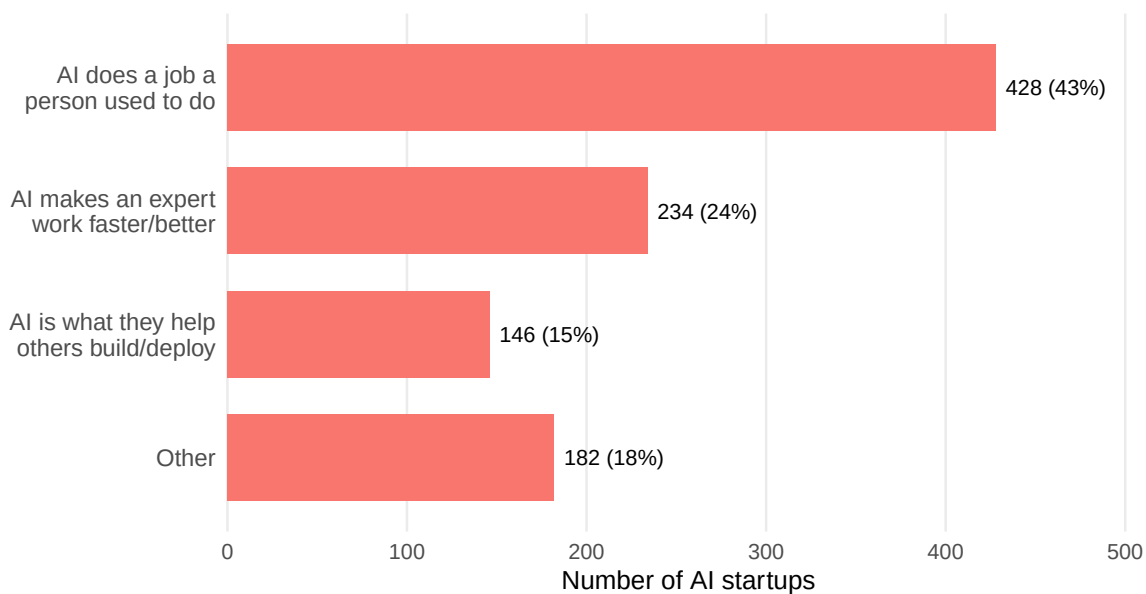


Figure 7: Classification of 990 AI-tagged Y Combinator startups (batches W20–F24) based on company descriptions. An LLM read each startup’s description and classified it into one of four inductively derived categories. Roughly 67% of AI startups embed AI directly in their products using it to automate or augment workers at other firms.

## 9 Tables

Table 1: Summary Statistics

| <i>Panel A: Y Combinator Sample</i> |                     |        |        |                 |         |       |
|-------------------------------------|---------------------|--------|--------|-----------------|---------|-------|
| Variable                            | Non-AI Startups     |        |        | AI Startups     |         |       |
|                                     | Mean                | SD     | N      | Mean            | SD      | N     |
| Employees (Revelio)                 | 42.30               | 182.97 | 1,479  | 12.64           | 22.73   | 754   |
| Hierarchy Layers                    | 5.01                | 1.93   | 1,479  | 3.76            | 1.97    | 754   |
| Manager Share                       | 0.242               | 0.189  | 1,479  | 0.174           | 0.185   | 754   |
| Entry-Level Share (Sen 1-2)         | 0.300               | 0.214  | 1,479  | 0.190           | 0.208   | 754   |
| Senior Share (Sen 6-7)              | 0.277               | 0.263  | 1,479  | 0.449           | 0.329   | 754   |
| Engineer Share                      | 0.358               | 0.245  | 1,479  | 0.449           | 0.276   | 754   |
| Sales Share                         | 0.226               | 0.186  | 1,479  | 0.169           | 0.184   | 754   |
| Total Raised (\$M)                  | 12.49               | 63.10  | 1,471  | 4.07            | 12.33   | 747   |
| Valuation (\$M)                     | 55.44               | 293.83 | 796    | 25.92           | 57.73   | 250   |
| Funding/Emp (\$M)                   | 0.37                | 0.64   | 1,471  | 0.37            | 0.71    | 747   |
| Valuation/Emp (\$M)                 | 1.22                | 2.15   | 796    | 1.96            | 3.69    | 250   |
| <i>Panel B: PitchBook Sample</i>    |                     |        |        |                 |         |       |
| Variable                            | Non-AI (prob < 0.5) |        |        | AI (prob ≥ 0.5) |         |       |
|                                     | Mean                | SD     | N      | Mean            | SD      | N     |
| Employees (Revelio)                 | 20.27               | 76.73  | 36,087 | 17.46           | 43.32   | 5,127 |
| Hierarchy Layers                    | 4.43                | 1.97   | 36,087 | 4.45            | 1.92    | 5,127 |
| Manager Share                       | 0.337               | 0.249  | 36,087 | 0.268           | 0.218   | 5,127 |
| Entry-Level Share (Sen 1-2)         | 0.271               | 0.225  | 36,087 | 0.237           | 0.206   | 5,127 |
| Senior Share (Sen 6-7)              | 0.329               | 0.277  | 36,087 | 0.358           | 0.271   | 5,127 |
| Engineer Share                      | 0.256               | 0.239  | 36,087 | 0.385           | 0.244   | 5,127 |
| Sales Share                         | 0.230               | 0.218  | 36,087 | 0.205           | 0.196   | 5,127 |
| Total Raised (\$M)                  | 11.33               | 75.86  | 25,640 | 18.08           | 351.84  | 3,696 |
| Valuation (\$M)                     | 47.34               | 329.19 | 11,294 | 157.66          | 2041.81 | 1,597 |
| Funding/Emp (\$M)                   | 0.60                | 4.19   | 25,640 | 0.51            | 4.08    | 3,696 |
| Valuation/Emp (\$M)                 | 2.37                | 14.68  | 11,294 | 2.77            | 14.79   | 1,597 |

*Notes:* Panel A presents summary statistics for Y Combinator startups from batches W20–F24. AI status is determined by YC tags containing “artificial-intelligence” or “ai.” Panel B presents summary statistics for PitchBook startups with first financing deal in 2020–2024. AI probability is predicted using a Random Forest classifier trained on YC data; “AI (prob ≥ 0.5)” indicates predicted probability ≥ 0.5. Employee data from Revelio Labs as of January 2025. Financial data from PitchBook.



Table 2: AI and Firm Size

|                                        | Y Combinator Sample  |                      |                      | PitchBook Sample     |                      |                      |
|----------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                        | (1)<br>Levels        | (2)<br>Log           | (3)<br>Poisson       | (4)<br>Levels        | (5)<br>Log           | (6)<br>Poisson       |
| AI Startup                             | -8.074***<br>(1.909) | -0.282***<br>(0.052) | -0.480***<br>(0.104) | -6.437***<br>(1.524) | -0.128***<br>(0.037) | -0.409***<br>(0.100) |
| Non-AI Mean                            | 33.78                |                      |                      | 17.83                |                      |                      |
| Observations                           | 2,786                | 2,233                | 2,771                | 47,007               | 41,214               | 47,003               |
| R <sup>2</sup> / Pseudo-R <sup>2</sup> | 0.272                | 0.284                |                      | 0.029                | 0.049                |                      |
| Time × Industry FE                     | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |

*Notes:* This table reports estimates of the relationship between AI status and firm size. The dependent variable is employee count from Revelio Labs as of January 2025. Columns (1)–(3) use Y Combinator startups from batches W20–F24 with AI as a binary indicator from YC tags. Columns (4)–(6) use PitchBook startups with first deal in 2020–2024, with AI as an estimated probability (0–1) from a Random Forest classifier trained on YC data. Log specifications exclude firms with zero employees. All specifications include batch × industry (YC) or deal quarter × industry (PB) fixed effects. Sample sizes: YC level uses 2,786 firms; YC Poisson reports 2,771 (firms in singleton fixed-effect cells are dropped); YC log uses 2,233 firms (those with at least one Revelio-matched employee). PitchBook level uses 47,007 firms; PB Poisson reports 47,003 (again due to dropped singleton cells); PB log uses 41,214 firms. The Non-AI Mean row is computed on the level-regression sample with zeros imputed for firms without a Revelio match; restricting to Revelio-matched firms (as in Table 1) yields a higher non-AI mean (42.30 in YC; 20.27 in PB). HC1 robust standard errors in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 3: AI and Functional Workforce Composition

| <i>Panel A: Y Combinator Sample</i> |                     |                     |                      |                      |                      |                      |                      |
|-------------------------------------|---------------------|---------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                     | (1)<br>Engineer     | (2)<br>Scientist    | (3)<br>Sales         | (4)<br>Marketing     | (5)<br>Operations    | (6)<br>Finance       | (7)<br>Admin         |
| AI Startup                          | 0.045***<br>(0.013) | 0.025***<br>(0.008) | -0.021**<br>(0.009)  | -0.012<br>(0.009)    | -0.009**<br>(0.004)  | -0.017***<br>(0.005) | -0.011**<br>(0.006)  |
| Non-AI Mean                         | 0.358               | 0.080               | 0.226                | 0.145                | 0.052                | 0.059                | 0.080                |
| Observations                        | 2,233               | 2,233               | 2,233                | 2,233                | 2,233                | 2,233                | 2,233                |
| R <sup>2</sup>                      | 0.126               | 0.399               | 0.085                | 0.023                | 0.055                | 0.031                | 0.030                |
| Batch $\times$ Industry FE          | Yes                 | Yes                 | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| <i>Panel B: PitchBook Sample</i>    |                     |                     |                      |                      |                      |                      |                      |
|                                     | (1)<br>Engineer     | (2)<br>Scientist    | (3)<br>Sales         | (4)<br>Marketing     | (5)<br>Operations    | (6)<br>Finance       | (7)<br>Admin         |
| AI Startup                          | 0.183***<br>(0.008) | 0.067***<br>(0.005) | -0.056***<br>(0.007) | -0.052***<br>(0.006) | -0.049***<br>(0.004) | -0.070***<br>(0.003) | -0.022***<br>(0.004) |
| Non-AI Mean                         | 0.256               | 0.096               | 0.230                | 0.187                | 0.084                | 0.062                | 0.085                |
| Observations                        | 41,214              | 41,214              | 41,214               | 41,214               | 41,214               | 41,214               | 41,214               |
| R <sup>2</sup>                      | 0.150               | 0.339               | 0.047                | 0.087                | 0.024                | 0.060                | 0.013                |
| Deal Qtr $\times$ Industry FE       | Yes                 | Yes                 | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |

*Notes:* This table reports estimates of the relationship between AI status and workforce functional composition. The dependent variables are the share of employees in each functional category based on Revelio Labs job classifications. For YC, AI is a binary indicator from tags; for PitchBook, AI is an estimated probability (0–1) from a Random Forest classifier. All specifications include batch  $\times$  industry (YC) or deal quarter  $\times$  industry (PB) fixed effects. Sample: 2,233 YC startups and 41,214 PitchBook firms with at least one employee. HC1 robust standard errors in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 4: AI and Workforce Seniority Composition

|                           | Y Combinator Sample  |                   |                     | PitchBook Sample     |                  |                     |
|---------------------------|----------------------|-------------------|---------------------|----------------------|------------------|---------------------|
|                           | (1)<br>Entry (1-2)   | (2)<br>Mid (3-5)  | (3)<br>Senior (6-7) | (4)<br>Entry (1-2)   | (5)<br>Mid (3-5) | (6)<br>Senior (6-7) |
| AI Startup                | -0.040***<br>(0.010) | -0.012<br>(0.012) | 0.052***<br>(0.013) | -0.060***<br>(0.007) | 0.004<br>(0.008) | 0.056***<br>(0.009) |
| Non-AI Mean               | 0.300                | 0.424             | 0.277               | 0.271                | 0.400            | 0.329               |
| Observations              | 2,233                | 2,233             | 2,233               | 41,214               | 41,214           | 41,214              |
| R <sup>2</sup>            | 0.194                | 0.111             | 0.314               | 0.030                | 0.016            | 0.041               |
| Time $\times$ Industry FE | Yes                  | Yes               | Yes                 | Yes                  | Yes              | Yes                 |

*Notes:* This table reports estimates of the relationship between AI status and workforce seniority composition. The dependent variables are the share of employees at each seniority level based on Revelio Labs' 1–7 seniority scale: Entry-level (1–2), Mid-level (3–5), and Senior (6–7). For YC, AI is a binary indicator from tags; for PitchBook, AI is an estimated probability (0–1) from a Random Forest classifier. All specifications include batch  $\times$  industry (YC) or deal quarter  $\times$  industry (PB) fixed effects. Sample: 2,233 YC startups and 41,214 PitchBook firms with at least one employee. HC1 robust standard errors in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 5: AI and Organizational Hierarchy

|                           | Y Combinator Sample  |                      | PitchBook Sample    |                      |
|---------------------------|----------------------|----------------------|---------------------|----------------------|
|                           | (1)<br>Layers        | (2)<br>Mgr Share     | (3)<br>Layers       | (4)<br>Mgr Share     |
| AI Startup                | -0.482***<br>(0.089) | -0.037***<br>(0.010) | -0.146**<br>(0.063) | -0.112***<br>(0.007) |
| Non-AI Mean               | 5.01                 | 0.242                | 4.43                | 0.337                |
| Observations              | 2,233                | 2,233                | 41,214              | 41,214               |
| R <sup>2</sup>            | 0.270                | 0.064                | 0.039               | 0.038                |
| Time $\times$ Industry FE | Yes                  | Yes                  | Yes                 | Yes                  |

*Notes:* This table reports estimates of the relationship between AI status and organizational hierarchy. “Layers” is the count of distinct seniority levels (1–7) present in the firm. “Mgr Share” is the fraction of employees in management occupations (O\*NET codes starting with 11-). For YC, AI is a binary indicator from tags; for PitchBook, AI is an estimated probability (0–1) from a Random Forest classifier. All specifications include batch  $\times$  industry (YC) or deal quarter  $\times$  industry (PB) fixed effects. Sample restricted to firms with at least one Revelio-matched employee: 2,233 YC startups and 41,214 PitchBook firms. HC1 robust standard errors in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 6: AI and Startup Financing

| <i>Panel A: Total Financing</i>         |                     |                      |                     |                      |
|-----------------------------------------|---------------------|----------------------|---------------------|----------------------|
|                                         | Y Combinator Sample |                      | PitchBook Sample    |                      |
|                                         | (1)                 | (2)                  | (3)                 | (4)                  |
|                                         | Log Funding         | Log Valuation        | Log Funding         | Log Valuation        |
| AI Startup                              | -0.165**<br>(0.067) | -0.075<br>(0.118)    | 0.136*<br>(0.070)   | 0.458***<br>(0.100)  |
| Observations                            | 2,765               | 1,275                | 33,309              | 14,413               |
| R <sup>2</sup>                          | 0.100               | 0.039                | 0.080               | 0.053                |
| Time × Industry FE                      | Yes                 | Yes                  | Yes                 | Yes                  |
| <i>Panel B: Per-Employee Efficiency</i> |                     |                      |                     |                      |
|                                         | Y Combinator Sample |                      | PitchBook Sample    |                      |
|                                         | (1)                 | (2)                  | (3)                 | (4)                  |
|                                         | Log<br>Funding/Emp  | Log<br>Valuation/Emp | Log<br>Funding/Emp  | Log<br>Valuation/Emp |
| AI Startup                              | 0.109*<br>(0.063)   | 0.260**<br>(0.131)   | 0.285***<br>(0.063) | 0.564***<br>(0.089)  |
| Non-AI Mean (\$M)                       | 0.37                | 1.22                 | 0.60                | 2.37                 |
| Observations                            | 2,218               | 1,046                | 29,336              | 12,891               |
| R <sup>2</sup>                          | 0.057               | 0.131                | 0.071               | 0.055                |
| Time × Industry FE                      | Yes                 | Yes                  | Yes                 | Yes                  |

*Notes:* Panel A reports estimates of the relationship between AI status and total startup financing. Panel B reports estimates of per-employee efficiency. The dependent variables are log total funding raised, log last known valuation (Panel A), and their per-employee equivalents (Panel B), using PitchBook financial data and Revelio Labs employee counts. For YC, AI is a binary indicator from tags; for PitchBook, AI is an estimated probability (0–1) from a Random Forest classifier. All specifications include batch × industry (YC) or deal quarter × industry (PB) fixed effects. HC1 robust standard errors in parentheses. Sample size differences are driven by variation in funding and valuation coverage in our PitchBook data. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 7: How AI-Native Startups Embed AI in Their Products

| Company                                                                           | How AI Creates Value                                                                                                                                                                     |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Autonomous Workflow: AI does a job a person used to do (N = 428, 43%)</b>      |                                                                                                                                                                                          |
| Artisan (W24)                                                                     | AI autonomously performs outbound sales development by discovering leads, researching prospects, and writing personalized emails.                                                        |
| Aurelian (S22)                                                                    | AI acts as an automated phone operator for 911 centers, instantly routing, triaging, and resolving non-emergency calls.                                                                  |
| Flagright (W22)                                                                   | AI-native AML compliance platform: generative AI automates transaction monitoring, screening, investigation, and reporting for fintechs and banks.                                       |
| Scratch (W24)                                                                     | AI agents automate veterinary clinic operations including scheduling and clinical workflows, handling front-desk tasks.                                                                  |
| FazeShift (S24)                                                                   | AI automates the entire accounts receivable workflow, connecting fragmented data across platforms and replacing teams doing manual AR work.                                              |
| <b>Expert Augmentation: AI makes an expert work faster/better (N = 234, 24%)</b>  |                                                                                                                                                                                          |
| Tower (W24)                                                                       | AI accelerates legal due diligence by helping lawyers stay organized, spot issues earlier, and close deals faster.                                                                       |
| Agentic Labs (W24)                                                                | AI connects to an engineering codebase to help developers write technical design documents faster.                                                                                       |
| Medium Biosciences (S21)                                                          | AI models design functional therapeutic proteins, augmenting biologists in protein engineering and drug development.                                                                     |
| OffDeal (W24)                                                                     | AI-native investment bank: AI automates analyst-level deal analysis and documentation, enabling M&A advisors to focus on strategic client interactions.                                  |
| Iliad (W23)                                                                       | AI generates images and 3D models with granular artistic control, augmenting artists and designers.                                                                                      |
| <b>AI Infrastructure: AI is what they help others build/deploy (N = 146, 15%)</b> |                                                                                                                                                                                          |
| Roboflow (S20)                                                                    | AI-powered computer vision platform that trains and deploys object detection and image classification models, sold as tools for developers to add visual intelligence to their software. |
| Exa (S21)                                                                         | AI-powered search engine rebuilt from scratch using embeddings and neural retrieval, sold as an API so developers can bring web knowledge into LLM-powered products.                     |
| Tavus (S21)                                                                       | AI generates photorealistic video avatars and real-time conversational video via API, enabling developers to create AI employees and assistants with a human face.                       |
| Firecrawl (S22)                                                                   | AI-powered web scraping that converts any URL into clean, LLM-ready markdown or structured data via a single API call, serving as a data pipeline for AI application builders.           |
| Encord (W21)                                                                      | AI-assisted data labeling platform where models pre-annotate images and video, with humans reviewing corrections—helping AI teams get vision models into production faster.              |
| <b>Other (N = 182, 18%)</b>                                                       |                                                                                                                                                                                          |
| Reflex (W23)                                                                      | Open-source Python web framework; AI agents mentioned but the core product is a web development framework.                                                                               |
| Campfire (W22)                                                                    | AI agents powered by LLMs serve as core gameplay characters and drive interactive game experiences, creating AI-native entertainment.                                                    |
| K-Scale Labs (W24)                                                                | AI powers open-source humanoid robots that walk, talk, and manipulate objects, straddling hardware and AI.                                                                               |
| Pointhound (S23)                                                                  | Helps users earn and redeem credit card points for flights; no AI functionality described.                                                                                               |
| Envelope (S23)                                                                    | Banking app with built-in budgeting; the description does not mention AI as part of the product.                                                                                         |

*Notes:* An LLM read each startup’s description and wrote a one-sentence characterization of how AI creates value, then classified into one of four categories. Categories were derived inductively from qualitative reading of a random sample of descriptions.

Table 8: AI Product and AI Process: Effects on Firm Size and Hierarchy

|                                        | (1)<br>Levels      | (2)<br>Log           | (3)<br>Poisson       | (4)<br>Layers       | (5)<br>Mgr Share  |
|----------------------------------------|--------------------|----------------------|----------------------|---------------------|-------------------|
| AI Product                             | -10.16**<br>(4.16) | -0.144***<br>(0.049) | -0.212***<br>(0.081) | -0.211**<br>(0.085) | 0.003<br>(0.010)  |
| AI Process                             | 45.21*<br>(26.18)  | 0.092<br>(0.083)     | 0.204<br>(0.161)     | 0.039<br>(0.121)    | -0.020<br>(0.014) |
| Observations                           | 1,684              | 1,666                | 1,684                | 1,666               | 1,666             |
| R <sup>2</sup> / Pseudo-R <sup>2</sup> | 0.404              | 0.584                |                      | 0.438               | 0.140             |
| Log Postings control                   | Yes                | Yes                  | Yes                  | Yes                 | Yes               |
| Batch $\times$ Industry FE             | Yes                | Yes                  | Yes                  | Yes                 | Yes               |

*Notes:* This table reports estimates of the relationship between AI Product, AI Process, firm size, and hierarchy. Sample: YC startups in batches W20–F24 with at least one non-staffing Revelio job posting on or before January 31, 2025. AI Product is a binary indicator equal to one if the firm’s YC description fits the “AI does a job a person used to do” or “AI makes an expert work faster/better” categories from the GPT-4.1 taxonomy (Section 5). AI Process is a binary indicator equal to one if any non-staffing posting names an AI tool from our 33-entry dictionary (e.g., Cursor, GitHub Copilot, ChatGPT, Claude; full list in Appendix). Columns (1) and (3) use the full posting sample ( $N = 1,684$ ); columns (2), (4), and (5) drop firms with zero Revelio-matched employees ( $N = 1,666$ ). All specifications include the log count of non-staffing postings as a labor-demand control and batch  $\times$  industry fixed effects. HC1 robust standard errors in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 9: AI Treatment-Effect Heterogeneity by Industry Group: Employment and Hierarchy

|                                        | (1)<br>Levels       | (2)<br>Log           | (3)<br>Poisson       | (4)<br>Layers        | (5)<br>Mgr Share     |
|----------------------------------------|---------------------|----------------------|----------------------|----------------------|----------------------|
| AI Startup                             | -4.45***<br>(1.68)  | -0.207***<br>(0.060) | -0.286***<br>(0.108) | -0.375***<br>(0.104) | -0.045***<br>(0.011) |
| AI $\times$ Services-delivery          | -23.39***<br>(8.97) | -0.410***<br>(0.141) | -0.929***<br>(0.274) | -0.450**<br>(0.226)  | 0.021<br>(0.024)     |
| AI $\times$ Other                      | -0.56<br>(3.07)     | -0.156<br>(0.155)    | -0.101<br>(0.222)    | -0.489*<br>(0.286)   | 0.052*<br>(0.030)    |
| Observations                           | 2,786               | 2,233                | 2,771                | 2,233                | 2,233                |
| R <sup>2</sup> / Pseudo-R <sup>2</sup> | 0.323               | 0.345                |                      | 0.332                | 0.141                |
| Batch $\times$ Industry FE             | Yes                 | Yes                  | Yes                  | Yes                  | Yes                  |

*Notes:* This table reports estimates of the relationship between AI status and firm size and hierarchy interacted with industry-group bucket. Sample: YC startups in batches W20–F24 with non-missing IndustryGroup. AI is a binary indicator from YC tags. Industry-Groups are collapsed into three buckets: *Software* (reference); *services-delivery* (Healthcare Services, Services (Non-Financial), Healthcare Technology Systems, Commercial Services, Media, Healthcare, and Other Healthcare); *Other* (all remaining IndustryGroups). The implied AI effect in services-delivery and Other is the sum of the AI Startup row and the corresponding interaction row. Columns (1) and (3) use the full sample; columns (2), (4), and (5) drop firms with zero Revelio-matched employees. All specifications include batch  $\times$  industry fixed effects (matching Tables 2–3). HC1 robust standard errors in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .



## A Appendix

This appendix provides supporting material in four sections, followed by all appendix figures and tables collected at the end. Section A.1 documents the sample-construction funnel. The next two sections describe how we measure the two channels: §A.2 the AI value classification methodology, and §A.3 the process-channel measure and dictionary of 33 named AI tools. Section A.4 reviews existing industry reports on AI startups. Figures A1–A2 illustrate how AI status is determined from YC tags. Figure A3 and Figure A4 present firm size distributions, with Table A4 showing robustness checks. Table A5 and Figure A5 address hierarchy analysis. Figure A6 shows valuation per employee distributions. Tables A1–A2 summarize industry reports on AI startups. Table A3 consolidates the sample-construction funnel, field-coverage rates, and per-batch Revelio match rates. Tables A6–A8 examine workforce demographics, education, and geography. Tables A9–A11 replicate these analyses using the broader PitchBook sample.

### A.1 Sample Construction and Filtering

Table A3 documents the funnel of filters that takes us from the raw YC and PitchBook rosters to the analytical samples used throughout the paper, and identifies the four canonical sample sizes (YC regression sample, YC workforce sample, PitchBook regression sample, PitchBook workforce sample) that all subsequent N’s trace back to. Per-row N values in individual regression tables can be smaller than the canonical numbers when a dependent variable has additional missingness; those cases are flagged in the corresponding table notes.

### A.2 AI Value Classification Methodology

To classify how AI creates value in AI-native startups (Figure 7 and Table 7), we use an LLM-based classification approach. Figure 7 and Table 7 use an AI-tagged-only pass of this schema over the 990 AI-tagged W20–F24 YC firms. For each firm, the LLM reads the startup’s company name, batch, industry, one-liner, and full description and produces two outputs: (1) a one-sentence characterization of the core way AI creates value for the firm, and (2) a classification into exactly one of four categories. To construct the AI Product indicator used in Table 8 and in the base rates in Section 5, we re-apply the same schema to all 2,891 W20–F24 YC firms (AI- and non-AI-tagged) using GPT-4.1 with structured outputs, so the product-channel measure is defined for the full bilateral sample.

**Category Derivation.** Categories were derived inductively. We drew a random sample of 40 AI startup descriptions (seed = 99) and, for each, wrote a free-form one-sentence characterization of how AI creates value—with no predefined categories. Natural patterns emerged from this qualitative reading, which we formalized into four categories:

1. *AI does a job a person used to do*—The AI performs a complete role or task end-to-end autonomously. Examples: AI sales representative, AI support agent, AI recruiter, AI bookkeeper, AI content writer. The AI *is* the worker.
2. *AI makes an expert work faster/better*—The AI is a tool within a professional’s workflow, augmenting their expertise. Examples: AI that helps lawyers review documents, helps engineers design parts, helps doctors diagnose, helps scientists discover drugs. The human is still doing the job; AI makes them dramatically more productive.
3. *AI is what they help others build/deploy*—Infrastructure, platforms, and tools for other people to build or use AI. Examples: LLM evaluation platforms, API gateways, AI training tools, vector databases, fine-tuning infrastructure. Picks and shovels for the AI economy.
4. *Other*—No clear AI role in the company description, AI tag appears to be noise, or the company does not fit the above categories.

**Classification Process.** For our product and process regressions, we used GPT-4.1 to classify all firms in our YC data to generate product measures both for those tagged as AI startups and for firms without the tag. The four categories and guidelines are otherwise identical to the inductive derivation above. The LLM was given the following prompt:

*You are classifying YC startups. For EACH company, produce:*

1. **AI value sentence:** *One sentence describing the core way AI creates value for this firm. Be concrete and specific. Focus on what the AI actually does in their product—is it automating a task? Analyzing data? Generating content? Building tools for others?*

2. **Category:** *Exactly one of these four:*

- “AI does a job a person used to do”—The AI performs a complete role/task end-to-end autonomously (e.g., AI sales rep, AI support agent, AI recruiter, AI bookkeeper)
- “AI makes an expert work faster/better”—The AI is a tool within a professional’s workflow, augmenting their expertise (e.g., AI helps lawyers review docs, helps engineers design, helps doctors diagnose)

- *“AI is what they help others build/deploy”—Infrastructure/platforms/tools for other people to build or use AI (e.g., LLM eval platforms, API gateways, AI training tools, vector databases)*
- *“Other”—No clear AI role in the description, AI tag seems like noise, or doesn’t fit the above categories*

**IMPORTANT GUIDELINES:**

- *“Creates something new” (generative) should go into one of the first two categories depending on context: writing sales emails = doing a job; helping a musician create sounds = making an expert better*
- *If the description doesn’t mention AI at all but the company is tagged AI, classify as “Other”*
- *Be grounded in what the description actually says, not what you imagine the company might do*

### A.3 Process-Channel Measure: Named AI Tools Dictionary

The process-channel indicator described in Section 5 is constructed from a curated dictionary of 33 named Gen AI assistance tools. For each non-staffing Revelio Labs posting from a W20–F24 YC firm dated on or before January 31, 2025, we HTML-strip the description, lowercase it, and test for case-insensitive, word-boundary regex matches against each entry in the dictionary. A firm is coded as using AI in its internal process if at least one of its non-staffing postings matches at least one pattern. The dictionary was assembled by combining a corpus audit of named-AI-tool mentions in our window with the Stack Overflow 2024 Developer Survey AI-tools section and 2024 industry roundups of AI coding and workflow tools. All listed tools launched before January 2025. We deliberately exclude Hugging Face: a corpus audit showed its mentions are virtually all product-channel (workers being asked to *build with* HF transformers and APIs) rather than worker tool use.

The dictionary is organized into two categories:

- *Coding and general-purpose AI assistants*: ChatGPT, GitHub Copilot, GPT-4 (incl. 4o), Claude, GPT-3.5, Cursor, Zed, Devin, Codeium, V0, Cline, Sourcegraph Cody, JetBrains AI, Microsoft Copilot, Poe.
- *Workflow and creative AI tools*: Midjourney, Stable Diffusion, ElevenLabs, DALL·E, Harvey AI, RunwayML, Adobe Firefly, Pika Labs, 11x, Granola, Mem.ai, Read.ai, Loom AI, Zapier AI, Rewind AI, Superhuman AI, Sora.

We chose a named-tools dictionary over freer-form patterns (e.g., “uses AI tools,” “AI-assisted coding”) for three reasons. First, named tools are unambiguous: when a posting says “We are heavy users of V0, Cursor, and other AI tools,” workers are explicitly describing their stack. Second, the dictionary is high-precision by construction: only postings that name a specific AI product trigger a match. Third, the dictionary is auditable: a reader can scan the patterns above and understand how the measure is defined without proprietary classifiers or human judgment. The Overlap (S24) and Fintool (W23) example postings illustrating the build-vs-use distinction now appear in Section 5 (§5.2) alongside the rest of the process-channel measurement discussion.

## A.4 Industry Reports on AI Startups

A handful of industry reports have offered descriptive statistics on AI startups, though typically without systematic non-AI comparison groups or formal sampling frames. We summarize the headline numbers here.

**AI startup performance in absolute terms.** The Lean AI Leaderboard reports an average revenue per employee of \$2.32M and an average valuation per employee of \$125M for its curated set of AI startups, with average team sizes of roughly 20 employees.<sup>10</sup> Bandhakavi (2025) estimates that AI companies raised \$110B in global venture capital with year-over-year growth of 62%, against a –12% contraction in broader tech VC, and the United States accounted for 74% of global AI VC.<sup>11</sup> Moore and Andrusko (2025) report that median enterprise AI startups reach \$0.7M in revenue by month six and \$2.1M by month twelve, with time to Series A of roughly nine months on a median pre-Series A raise of \$4M; consumer AI startups reach \$2.5M and \$4.2M at the same milestones, raising a median of \$8M before Series A.<sup>12</sup>

**AI vs. non-AI comparisons.** Floyd (2025) contrasts “AI Native” firms with “AI Adopter” firms on four metrics: ARR growth (100% vs. 23%), net dollar retention (132% vs. 108%), gross dollar retention (89% vs. 81%), and R&D budgets (56% vs. 28% of revenue).<sup>13</sup> Stripe (2025) reports that AI startups reach \$1M ARR in a median 11.5 months versus 15 months for SaaS peers, and \$5M ARR in 24 versus 37 months. Comparing AI cohorts within Stripe, firms launched in 2020–2023 reach \$1M ARR in 5 months versus 16 months for pre-2020 AI startups, and \$5M ARR in 13 versus 41 months. AI startups also expand internationally faster, reaching 55 countries by year one versus 25 for SaaS, and 104 versus 57 by year four.<sup>14</sup>

**Limitations of existing estimates.** These reports collectively suggest that AI startups grow revenue faster, scale internationally faster, and operate at higher revenue and valuation per employee than non-AI peers. However, they typically lack matched comparison samples, rely on self-selected respondent pools (e.g., Stripe customers, Lean AI Leaderboard nominees), and do not condition on

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<sup>10</sup><https://leanaileaderboard.com/>

<sup>11</sup>Bandhakavi, “AI companies raised \$52bn in global VC funding in Q1 2025,” *TechMonitor*, April 2025 (<https://www.techmonitor.ai/digital-economy/ai-and-automation/ai-companies-52bn-global-vc-funding-q1-2025>).

<sup>12</sup>Moore & Andrusko, “What ‘Working’ Means in the Era of AI Apps,” a16z, 2025 (<https://a16z.com/revenue-benchmarks-ai-apps/>).

<sup>13</sup>Floyd, “Beyond Benchmarks 2025,” Emergence Capital (<https://www.emcap.com/beyond-benchmarks>).

<sup>14</sup>Stripe, “Inside the growth of the top AI companies on Stripe,” 2025 (<https://stripe.com/blog/inside-the-growth-of-the-top-ai-companies-on-stripe>).

cohort or industry. This motivates the systematic within-cohort, within-industry comparisons we conduct in the main text.

## Appendix Figures

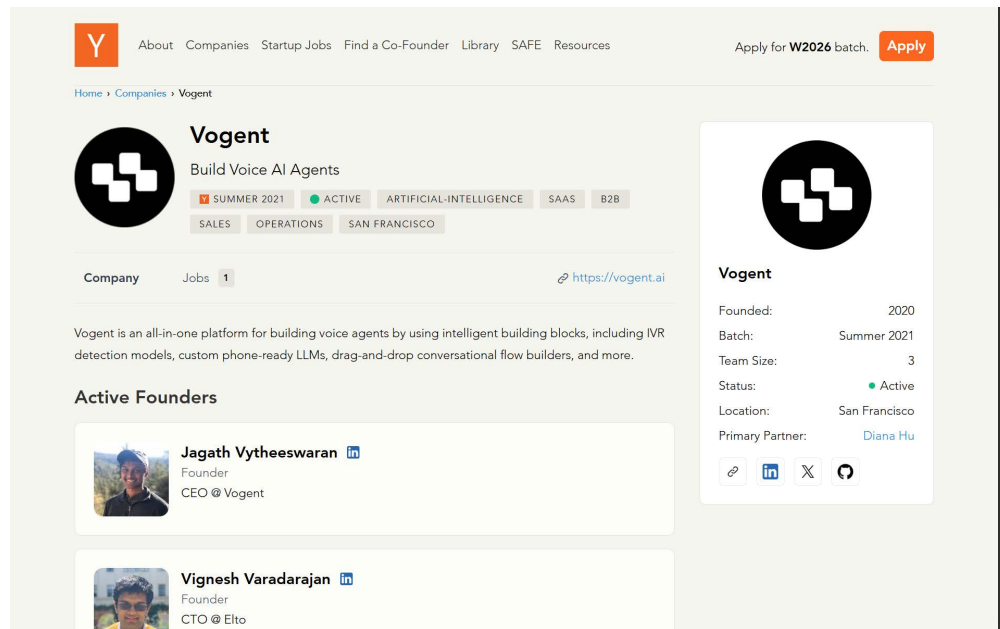


Figure A1: Screenshot from Y Combinator's webpage for Vogent, an AI startup, with the tags 'Artificial Intelligence', 'SAAS', 'B2B', 'Sales', and 'Operations'.

The screenshot shows the Y Combinator website for the startup Venostent. The header includes the Y Combinator logo and navigation links: About, Companies, Startup Jobs, Find a Co-Founder, Library, SAFE, and Resources. A button for 'Apply for W2026 batch' is visible in the top right.

The main content area for Venostent includes:

- Logo:** A stylized 'V' made of interlocking loops.
- Company Name:** Venostent
- Description:** Absorbable wrap for blood vessels
- Tags:** SUMMER 2020, ACTIVE, MEDICAL-DEVICES, 3D-PRINTING
- Navigation:** Company, Jobs (1), News, and a link to the website: <https://www.venostent.com>
- Description Text:** There are 10 million dialysis and bypass surgeries per year, but half of them fail. Venostent saves lives with a wrap that goes around blood vessels during surgery so they don't fail.
- Active Founders:**
  - Tim Boire** (Founder)
    - President and CEO of Venostent, Inc. I developed these novel smart polymeric biomaterials for vascular applications during my PhD at Vanderbilt University. Focus of the technology shifted to hemodialysis access applications after participating in the 2014 NSF I-Corps program. BS, Chemical & Biological Engineering and Biotechnology, Tufts University, 2008; PhD, Biomedical Engineering, Vanderbilt University, 2017. Worked in R&D at Genzyme Corporation, 2008 - 2011. Grew up in

A sidebar on the right provides additional details:

- VENOSTENT**
- Venostent**
- Founded:** 2017
- Batch:** Summer 2020
- Team Size:** 17
- Status:** Active
- Location:** Houston, TX
- Primary Partner:** Jared Friedman
- Social Links:** LinkedIn, Twitter, Facebook, and a generic link icon.

Figure A2: Screenshot from Y Combinator's webpage for Venostent, a non-AI startup, with the tags 'Medical-Devices' and '3D-Printing'.





Figure A3: Histograms of Employee Sizes by YC batch. Sample is the workforce sample that includes all YC firms from batches W20–F24 with at least one employee in the Revelio data,  $N = 2,233$  across all panels. Values winsorized at 50 employees ( $\approx 95$ th percentile firm size). Employee counts from Revelio Labs as of January 2025.

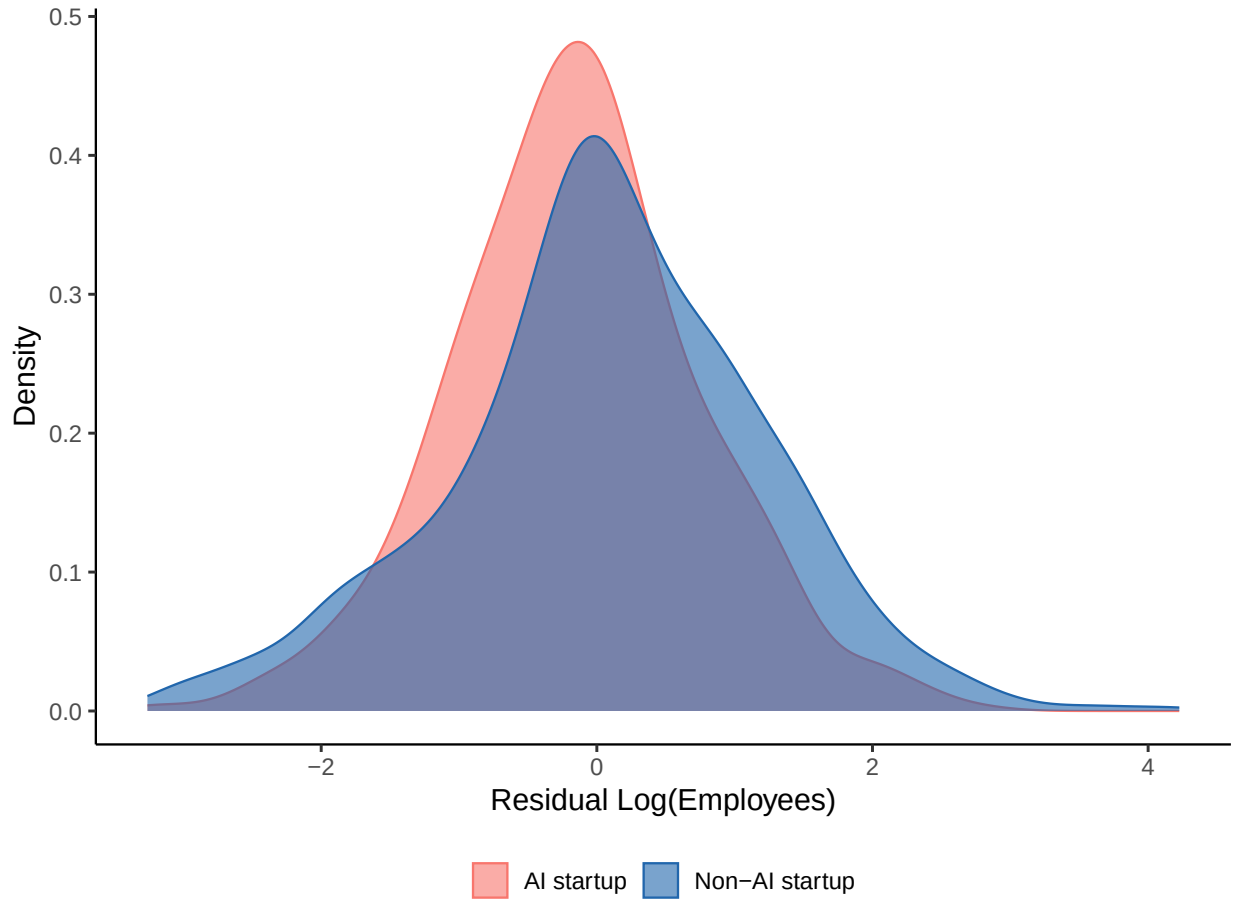


Figure A4: Kernel Density Plot of Firm Size Distribution. Distribution of log employee counts for AI (red) and non-AI (blue) Y Combinator startups, residualized with respect to batch  $\times$  industry fixed effects. Sample is the workforce sample that includes all YC firms from batches W20–F24 with at least one employee in the Revelio data,  $N = 2,233$ . Employee counts from Revelio Labs as of January 2025.

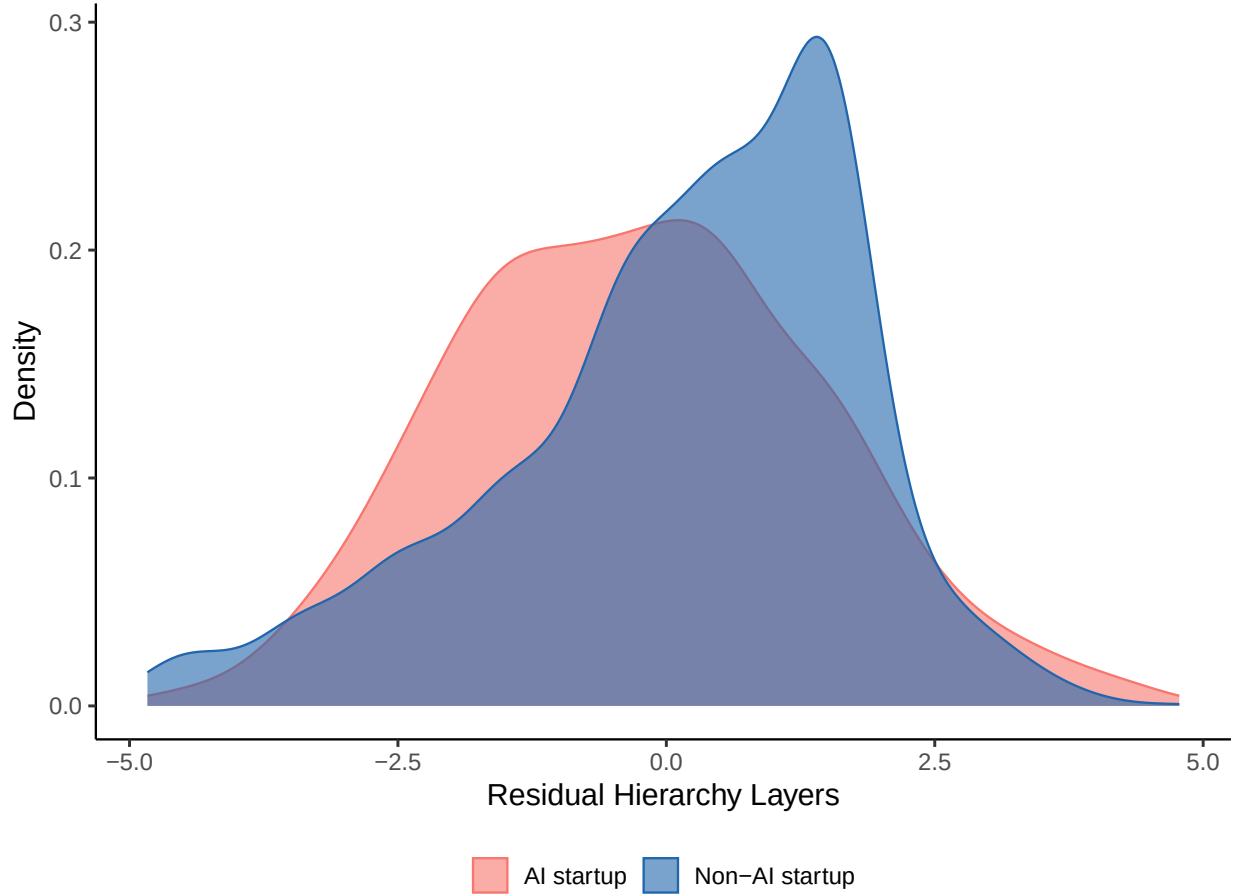


Figure A5: Kernel Density Plot of Hierarchy Depth Distribution. Distribution of hierarchy layers for AI (red) and non-AI (blue) Y Combinator startups, residualized with respect to batch  $\times$  industry fixed effects. AI startups are shifted leftward, indicating flatter organizational structures. Sample is the workforce sample that includes all YC firms from batches W20–F24 with at least one employee in the Revelio data,  $N = 2,233$ . Hierarchy layers measured as the count of distinct seniority levels (1–7) present in the firm from Revelio Labs as of January 2025.

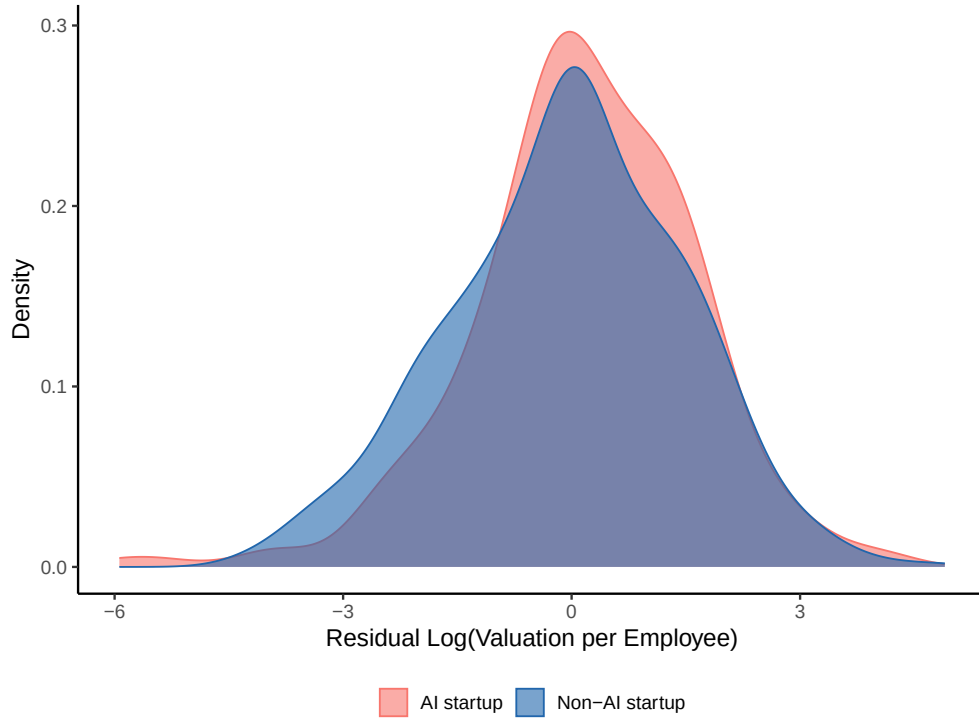


Figure A6: Kernel Density Plot of Valuation per Employee Distribution. Distribution of log valuation per employee for AI (red) and non-AI (blue) Y Combinator startups, residualized with respect to batch  $\times$  industry fixed effects. The rightward shift for AI startups indicates they achieve higher valuations per employee. Sample:  $N = 1,046$  YC firms from batches W20–F24 with both Revelio-matched employees and a non-missing PitchBook valuation. Firms with no recorded valuation are dropped rather than coded as zero. Valuation data from PitchBook; employee counts from Revelio Labs as of January 2025.

## Appendix Tables

Table A1: Literature Review: Statistics on AI Startups

| Study                   | Key Metrics                                                                                                                                                                                                                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lean AI Leaderboard     | Avg. revenue/employee: \$2.32M; Avg. valuation/employee: \$125M; Avg. team size: 20 employees                                                                                                                                                                                                      |
| Bandhakavi (2025)       | Global VC investment in AI: \$110B; YoY growth in AI VC: 62%; YoY growth in broader tech VC: -12%; US share of global AI VC: 74%                                                                                                                                                                   |
| Moore & Andrusko (2025) | <i>Enterprise AI</i> : Month 6 revenue median \$0.7M; Month 12 median \$2.1M; Time to Series A median 9 months; Pre-Series A raised median \$4.0M. <i>Consumer AI</i> : Month 6 revenue median \$2.5M; Month 12 median \$4.2M; Time to Series A median 8 months; Pre-Series A raised median \$8.0M |

*Notes:* This table summarizes quantitative metrics from recent literature on AI startup performance. Dollar amounts in millions (M) or billions (B) USD. Sources: Lean AI Leaderboard (<https://leanaileaderboard.com/>); Bandhakavi, “AI companies raised \$52bn in global VC funding in Q1 2025,” TechMonitor, April 2025 (<https://www.techmonitor.ai/digital-economy/ai-and-automation/ai-companies-52bn-global-vc-funding-q1-2025>); Moore & Andrusko, “What ‘Working’ Means in the Era of AI Apps,” a16z, 2025 (<https://a16z.com/revenue-benchmarks-ai-apps/>).

Table A2: Literature Review: Comparative Statistics on AI vs Non-AI Startups

| Study         | Key Metrics                                                                                                                                                                                                                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Floyd (2025)  | <i>AI Native</i> : ARR growth 100%, NDR 132%, GDR 89%, R&D budget 56%. <i>AI Adopters</i> : ARR growth 23%, NDR 108%, GDR 81%, R&D budget 28%                                                                                                                                                                                                             |
| Stripe (2025) | <i>Revenue milestones (AI vs. SaaS)</i> : Median months to \$1M ARR: 11.5 vs. 15; to \$5M ARR: 24 vs. 37. <i>AI cohort comparison (2020–23 vs. pre-2020)</i> : Months to \$1M ARR: 5 vs. 16; to \$5M ARR: 13 vs. 41. <i>International expansion (AI vs. SaaS)</i> : Year 1: 55 vs. 25 countries; Year 2: 79 vs. 42; Year 3: 91 vs. 51; Year 4: 104 vs. 57 |

*Notes:* This table summarizes comparative metrics from recent literature on AI vs. non-AI startup performance. ARR = Annual Recurring Revenue; NDR = Net Dollar Retention; GDR = Gross Dollar Retention. Dollar amounts in millions USD. Sources: Floyd, “Beyond Benchmarks 2025,” Emergence Capital (<https://www.emcap.com/beyond-benchmarks>); Stripe, “Inside the growth of the top AI companies on Stripe,” 2025 (<https://stripe.com/blog/inside-the-growth-of-the-top-ai-companies-on-stripe>).

Table A3: Sample Construction and Filtering

| Filter step                                                   | Firms (% of base) | Drop  |
|---------------------------------------------------------------|-------------------|-------|
| <i>Y Combinator sample (base: YC roster)</i>                  |                   |       |
| YC roster, batches W20–F24                                    | 2,891 (100.0%)    | —     |
| + PitchBook merge + non-missing IndustryGroup                 | 2,786 (96.4%)     | 105   |
| + Revelio-matched ( $\geq 1$ employee; workforce sample)      | 2,233 (77.2%)     | 553   |
| <i>Field coverage within YC roster:</i>                       |                   |       |
| Has funding data (totalraised > 0)                            | 2,765 (95.6%)     | —     |
| Has valuation data (lastknownvaluation > 0)                   | 1,275 (44.1%)     | —     |
| <i>PitchBook sample (base: 2020–2024 first-deal universe)</i> |                   |       |
| PitchBook firms, first deal year 2020–2024                    | 47,007            | —     |
| + Non-missing first deal date / quarter                       | 47,007            | 0     |
| + Non-missing IndustryGroup (regression sample)               | 47,007            | 0     |
| + Revelio-matched ( $\geq 1$ employee; workforce sample)      | 41,214            | 5,793 |

*Notes:* Each row in the YC and PitchBook funnels is an additive filter applied to the row above. For the YC block, the parenthetical % is firms as a share of the YC roster (N=2,891); field-coverage rows report how many YC roster firms have non-missing values in the indicated PitchBook field. The YC “regression sample” (used in Table 2) is the second YC funnel row (N=2,786). The YC “workforce sample” (Tables 1, 5, 4, 6, 3, and the firm-size and hierarchy figures) is the third YC funnel row (N=2,233). The PitchBook “regression sample” (Tables 2, 6, A11) is the third PB row (N=47,007); the PitchBook “workforce sample” (Tables 1, 5, 4, 6, 3, A9, and A10) is the fourth PB row (N=41,214). Revelio match rates range from 69.0% to 81.6% across YC batches and are nearly identical for AI and non-AI startups (76.2% vs. 77.8%).

Table A4: Firm Size Robustness: Comparing Data Sources (YC Sample Only)

|                           | YC Self-Reported    |                      | PitchBook            |                      | Revelio              |                      |
|---------------------------|---------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                           | (1)                 | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
|                           | Levels              | Log                  | Levels               | Log                  | Levels               | Log                  |
| AI Startup                | -5.714**<br>(2.746) | -0.147***<br>(0.041) | -9.113***<br>(2.754) | -0.233***<br>(0.042) | -8.074***<br>(1.909) | -0.282***<br>(0.052) |
| Non-AI Mean               | 25.61               |                      | 37.15                |                      | 33.78                |                      |
| Observations              | 2,786               | 2,730                | 2,786                | 2,668                | 2,786                | 2,233                |
| R <sup>2</sup>            | 0.028               | 0.270                | 0.120                | 0.297                | 0.272                | 0.284                |
| Time $\times$ Industry FE | Yes                 | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |

*Notes:* This table compares AI effects on firm size across three data sources for Y Combinator startups (batches W20–F24). Columns (1)–(2) use YC’s self-reported team size from application materials. Columns (3)–(4) use PitchBook employee counts. Columns (5)–(6) use Revelio Labs employee counts as of January 2025. Log specifications exclude firms with zero employees. All specifications include batch  $\times$  industry group fixed effects. The base sample is the canonical YC employment sample of 2,786 firms; per-column N values are smaller for log specifications and for sources with missing values. HC1 robust standard errors in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table A5: AI and Organizational Hierarchy: Size Controls

|                           | Y Combinator Sample   |                         | PitchBook Sample      |                         |
|---------------------------|-----------------------|-------------------------|-----------------------|-------------------------|
|                           | (1)<br>Layers         | (2)<br>Mgr Share        | (3)<br>Layers         | (4)<br>Mgr Share        |
| AI Startup                | -0.461***<br>(0.088)  | -0.038***<br>(0.010)    | -0.105*<br>(0.061)    | -0.113***<br>(0.007)    |
| N Employees               | 0.0021***<br>(0.0008) | -0.0001***<br>(0.00003) | 0.0062***<br>(0.0018) | -0.0002***<br>(0.00006) |
| Non-AI Mean               | 5.01                  | 0.242                   | 4.43                  | 0.337                   |
| Observations              | 2,233                 | 2,233                   | 41,214                | 41,214                  |
| R <sup>2</sup>            | 0.287                 | 0.065                   | 0.092                 | 0.042                   |
| Time $\times$ Industry FE | Yes                   | Yes                     | Yes                   | Yes                     |

*Notes:* This table reports estimates of the relationship between AI status and organizational hierarchy, controlling for firm size. “Layers” is the count of distinct seniority levels (1–7) present in the firm. “Mgr Share” is the fraction of employees in management occupations (O\*NET codes starting with 11-). For YC, AI is a binary indicator from tags; for PitchBook, AI is an estimated probability (0–1) from a Random Forest classifier. All specifications include batch  $\times$  industry (YC) or deal quarter  $\times$  industry (PB) fixed effects. Sample: 2,233 YC startups and 41,214 PitchBook firms. HC1 robust standard errors in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table A6: Summary Statistics: Demographic and Education Variables

| Variable                         | Mean   | SD      | Min | P25   | Median | P75    | Max  | % Zero | N    |
|----------------------------------|--------|---------|-----|-------|--------|--------|------|--------|------|
| <b>Panel A: Gender Counts</b>    |        |         |     |       |        |        |      |        |      |
| Female Employees                 | 7.582  | 30.636  | 0   | 0.000 | 2.000  | 6.000  | 954  | 27.5   | 1613 |
| Male Employees                   | 19.800 | 181.352 | 0   | 3.000 | 7.000  | 15.000 | 7183 | 1.2    | 1613 |
| <b>Panel B: Gender Shares</b>    |        |         |     |       |        |        |      |        |      |
| Female Share                     | 0.224  | 0.205   | 0   | 0.000 | 0.200  | 0.333  | 1    | 27.5   | 1613 |
| Male Share                       | 0.757  | 0.212   | 0   | 0.649 | 0.771  | 1.000  | 1    | 1.2    | 1613 |
| <b>Panel C: Ethnicity Counts</b> |        |         |     |       |        |        |      |        |      |
| White Employees                  | 10.534 | 21.715  | 0   | 2.000 | 4.000  | 11.000 | 313  | 9.3    | 1613 |
| Black Employees                  | 1.773  | 8.278   | 0   | 0.000 | 0.000  | 1.000  | 213  | 61.4   | 1613 |
| Hispanic Employees               | 3.180  | 17.148  | 0   | 0.000 | 0.000  | 1.000  | 366  | 57.1   | 1613 |
| Asian/Pacific Islander Employees | 12.066 | 198.433 | 0   | 0.000 | 2.000  | 5.000  | 7884 | 25.4   | 1613 |
| Non-White-Asian Employees        | 4.954  | 19.439  | 0   | 0.000 | 1.000  | 3.000  | 369  | 41.7   | 1613 |
| <b>Panel D: Ethnicity Shares</b> |        |         |     |       |        |        |      |        |      |
| White Share                      | 0.540  | 0.314   | 0   | 0.286 | 0.571  | 0.800  | 1    | 9.3    | 1613 |
| Black Share                      | 0.062  | 0.144   | 0   | 0.000 | 0.000  | 0.067  | 1    | 61.4   | 1613 |
| Hispanic Share                   | 0.087  | 0.176   | 0   | 0.000 | 0.000  | 0.091  | 1    | 57.1   | 1613 |
| Asian/Pacific Islander Share     | 0.302  | 0.312   | 0   | 0.000 | 0.200  | 0.500  | 1    | 25.4   | 1613 |
| Non-White-Asian Share            | 0.149  | 0.218   | 0   | 0.000 | 0.071  | 0.200  | 1    | 41.7   | 1613 |
| <b>Panel E: Education Counts</b> |        |         |     |       |        |        |      |        |      |
| Doctor Employees                 | 0.901  | 2.309   | 0   | 0.000 | 0.000  | 1.000  | 31   | 67.2   | 1613 |
| Master Employees                 | 4.368  | 17.733  | 0   | 0.000 | 2.000  | 4.000  | 630  | 26.6   | 1613 |
| MBA Employees                    | 2.453  | 33.336  | 0   | 0.000 | 0.000  | 2.000  | 1332 | 50.7   | 1613 |
| Bachelor Employees               | 12.463 | 104.673 | 0   | 2.000 | 4.000  | 9.000  | 4105 | 10.0   | 1613 |
| Associate Employees              | 0.271  | 1.075   | 0   | 0.000 | 0.000  | 0.000  | 19   | 86.4   | 1613 |
| High School Employees            | 0.493  | 2.516   | 0   | 0.000 | 0.000  | 0.000  | 77   | 79.4   | 1613 |
| <b>Panel F: Education Shares</b> |        |         |     |       |        |        |      |        |      |
| Doctor Share                     | 0.066  | 0.156   | 0   | 0.000 | 0.000  | 0.048  | 1    | 67.2   | 1613 |
| Master Share                     | 0.206  | 0.212   | 0   | 0.000 | 0.167  | 0.312  | 1    | 26.6   | 1613 |
| MBA Share                        | 0.074  | 0.124   | 0   | 0.000 | 0.000  | 0.105  | 1    | 50.7   | 1613 |
| Bachelor Share                   | 0.452  | 0.264   | 0   | 0.273 | 0.462  | 0.619  | 1    | 10.0   | 1613 |
| Associate Share                  | 0.009  | 0.040   | 0   | 0.000 | 0.000  | 0.000  | 1    | 86.4   | 1613 |
| High School Share                | 0.016  | 0.050   | 0   | 0.000 | 0.000  | 0.000  | 0    | 79.4   | 1613 |
| <b>Panel G: Prestige</b>         |        |         |     |       |        |        |      |        |      |
| Average Prestige                 | 0.419  | 0.262   | -0  | 0.256 | 0.447  | 0.619  | 1    | 0.0    | 1613 |
| Maximum Prestige                 | 0.800  | 0.169   | -0  | 0.754 | 0.844  | 0.893  | 1    | 0.0    | 1613 |

*Notes:* Summary statistics for demographic and education variables, with firm-level Revelio data aggregated from employee-level records. Panels A–B report gender counts and shares; Panels C–D report ethnicity counts and shares; Panels E–F report counts and shares by highest degree; Panel G reports prestige scores. Sample: 1,613 YC firms with non-missing Revelio demographic and education data (a subset of the 2,233 workforce sample used elsewhere in the paper).



Table A7: Regression of Education Shares on Startup AI Status (YC Sample)

|                            | (1)<br>PhD Share | (2)<br>MA Share  | (3)<br>MBA Share | (4)<br>Bachelor Share |
|----------------------------|------------------|------------------|------------------|-----------------------|
| AI Startup                 | 1.64**<br>(0.78) | 3.13**<br>(1.26) | 0.23<br>(0.76)   | -0.18<br>(1.56)       |
| Non-AI Mean                | 6.0              | 18.8             | 7.5              | 44.8                  |
| Observations               | 1,613            | 1,613            | 1,613            | 1,613                 |
| R <sup>2</sup>             | 0.397            | 0.103            | 0.076            | 0.115                 |
| Batch $\times$ Industry FE | Yes              | Yes              | Yes              | Yes                   |

*Notes:* Dependent variables are workforce shares by highest degree, each multiplied by 100 for readability. AI is a binary indicator from YC tags. All specifications include batch  $\times$  IndustryGroup fixed effects. Sample: 1,613 YC firms with non-missing Revelio education data (a subset of the 2,233 workforce sample used elsewhere in the paper). HC1 robust standard errors in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table A8: Summary Statistics: Geographic Location Variables

| Variable      | AI Mean | Non-AI Mean | Difference | P-value | AI N | Non-AI N |
|---------------|---------|-------------|------------|---------|------|----------|
| San Francisco | 0.425   | 0.226       | 0.200      | <0.001  | 990  | 1,901    |
| California    | 0.537   | 0.353       | 0.184      | <0.001  | 990  | 1,901    |

*Notes:* Summary statistics for the headquarter locations of YC startups from PitchBook. P-values are two-sample t-tests comparing means between AI and non-AI firms. Sample: 2,891 YC firms with available HQ location data from batches W20–F24.

Table A9: PitchBook Sample: Regression of Employee Shares by Demographic Characteristics on AI Status

|                                         | (1)<br>Female        | (2)<br>White       | (3)<br>API          | (4)<br>Non-White-API |
|-----------------------------------------|----------------------|--------------------|---------------------|----------------------|
| AI Probability                          | -0.059***<br>(0.007) | 0.027**<br>(0.011) | 0.085***<br>(0.011) | -0.112***<br>(0.007) |
| Non-AI Mean                             | 0.300                | 0.632              | 0.226               | 0.142                |
| Observations                            | 41,214               | 41,214             | 41,214              | 41,214               |
| R <sup>2</sup>                          | 0.108                | 0.030              | 0.028               | 0.031                |
| First Deal Quarter $\times$ Industry FE | Yes                  | Yes                | Yes                 | Yes                  |

*Notes:* Dependent variables are shares of employees in each demographic group; API = Asian and Pacific Islander; Non-White-API = Black, Hispanic, or Native American. AI Probability is a continuous 0–1 measure from a Random Forest classifier trained on YC labels; coefficients represent the effect of moving from 0% to 100% AI probability. All specifications include first-deal-quarter  $\times$  IndustryGroup fixed effects. Sample: 41,214 VC-backed startups with first financing deal between 2020 and 2024 (inclusive) and Revelio Labs workforce data. HC1 robust standard errors in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table A10: PitchBook Sample: Regression of Education Shares and Prestige on AI Status

|                                         | (1)<br>PhD/MA/MBA   | (2)<br>Bachelor      | (3)<br>Sub-Bachelor  | (4)<br>Prestige     |
|-----------------------------------------|---------------------|----------------------|----------------------|---------------------|
| AI Probability                          | 0.176***<br>(0.008) | -0.125***<br>(0.010) | -0.010***<br>(0.002) | 0.219***<br>(0.009) |
| Non-AI Mean                             | 0.297               | 0.456                | 0.025                | 0.360               |
| Observations                            | 41,214              | 40,102               | 41,214               | 41,214              |
| R <sup>2</sup>                          | 0.151               | 0.103                | 0.025                | 0.103               |
| First Deal Quarter $\times$ Industry FE | Yes                 | Yes                  | Yes                  | Yes                 |

*Notes:* Columns (1)–(3) are workforce shares by highest degree attained (PhD/MA/MBA = Doctoral, Masters, or MBA; Sub-Bachelor = Associate or High School). Column (4) is the average university prestige score across employees (Revelio). AI Probability is a continuous 0–1 measure from a Random Forest classifier; coefficients represent the effect of moving from 0% to 100% AI probability. All specifications include first-deal-quarter  $\times$  IndustryGroup fixed effects. Sample: 41,214 VC-backed startups with first financing deal between 2020 and 2024 (inclusive) and Revelio Labs education data; the Bachelor column drops to 40,102 firms with a non-missing Bachelor share. HC1 robust standard errors in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table A11: PitchBook Sample: Geographic Location by AI Status

|                                         | (1)<br>San Francisco | (2)<br>California   |
|-----------------------------------------|----------------------|---------------------|
| AI Probability                          | 0.182***<br>(0.010)  | 0.290***<br>(0.012) |
| Non-AI Mean                             | 0.036                | 0.110               |
| Observations                            | 47,007               | 47,007              |
| R <sup>2</sup>                          | 0.047                | 0.043               |
| First Deal Quarter $\times$ Industry FE | Yes                  | Yes                 |

*Notes:* Dependent variables are 0–1 indicators for HQ location. AI Probability is a continuous 0–1 measure from a Random Forest classifier; coefficients represent the effect of moving from 0% to 100% AI probability. All specifications include first-deal-quarter  $\times$  IndustryGroup fixed effects. Sample: 47,007 VC-backed startups with first financing deal between 2020 and 2024 (inclusive) and available HQ location data. The sample is restricted to U.S.-based startups, so we report only San Francisco and California concentration. HC1 robust standard errors in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .